Manure Module

RuFaS Scientific Documentation

Manure

v.1

Part I
Manure Module

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I. Module Introduction

The responsibility of the manure module in RuFaS is to simulate the loss and/or gain of manure mass and nutrients on a daily timestep at each step of the manure management chain on a dairy farm. On the majority of farms, manure represents an important link in the cycling of nutrients through animals, land, crops, and back to animals. Therefore, modeling both greenhouse gas (GHG) emissions as well as non-GHG nutrient and losses is crucial in capturing nutrient flows through the whole-farm system. The manure module accomplishes its responsibilities by tracking a critical, core set of variables called "ManureStream" variables, which include key agronomic nutrients (N, P, K), carbon (e.g. total and volatile solids), water, ash, mass, and volume. The Manure Module receives manure excretion and bedding information from the animal module, then passes manure through the manure management chain, until manure is either applied to fields or exported. The specific steps of the manure management chain are user-defined, and the individual steps/options in manure management chains are referred to as "processors". The exact chemical, physical, or other processes that occur at each step along the management chain depend on what type of processor the manure is being held in. For example, parlor cleaning processors, which represent removing manure from the milking parlor and holding areas, principally add water to manure but do not estimate GH emissions or other nutrient losses. A slurry storage outdoor processor (a type of manure storage), however, estimates daily methane (CH_4) and ammonia (NH_3-N) losses, which are then reflected by reducing the quantity of nutrients remaining in the stored manure at the end of the day.

A. Structural setup of the Manure Module

Processors in the Manure Module fall into four basic classes, with multiple types available within each class. The processor classes are intended to represent the common, primary steps in manure management. They include manure handling (which refers to the daily activities of cleaning and removing manure from facilities or managing it in place), storage, and manure treatment, such as anaerobic digestion or mechanical solid liquid separation.

Table 1: Examples of processor types by class.

Classes	Types
Handler	Manual Scraping; Alley Scraper; Flush System; Parlor Cleaning
Digester	Continuous Mix
Separator	Screw Press; Rotary Screen
Storage	Composting; Open Lot; Bedded Pack; Slurry Storage Outdoor; Slurry Storage Underfloor; Anaerobic Lagoon

How manure moves between processors is based on defining the destination processor that any one processor should send its manure to, and the proportion(s) of manure from the originating processor that should go there. For example, if 50% of manure from a single pen is routed to storage A, and 50% to storage B, the manure handler processor assigned to that pen will have two destinations defined (storage A and storage B), with the proportion going to each defined as 0.50. By controlling destinations and proportions, the user is able to define aggregating (i.e., assigning manure from two separate processors to the same destination) and splitting (assigning manure from one processor to two or more locations) behavior. A visual example is below and illustrates the manure management chain without RuFaS-specific syntax. The next figure illustrates RuFaS-specific information (e.g. specific processor names, creation of manure streams in the animal module, etc.). Note that the splitting of manure generated from singular pens in the Animal Module illustrated in Figure 8 is covered in the Animal to Manure Connection document.

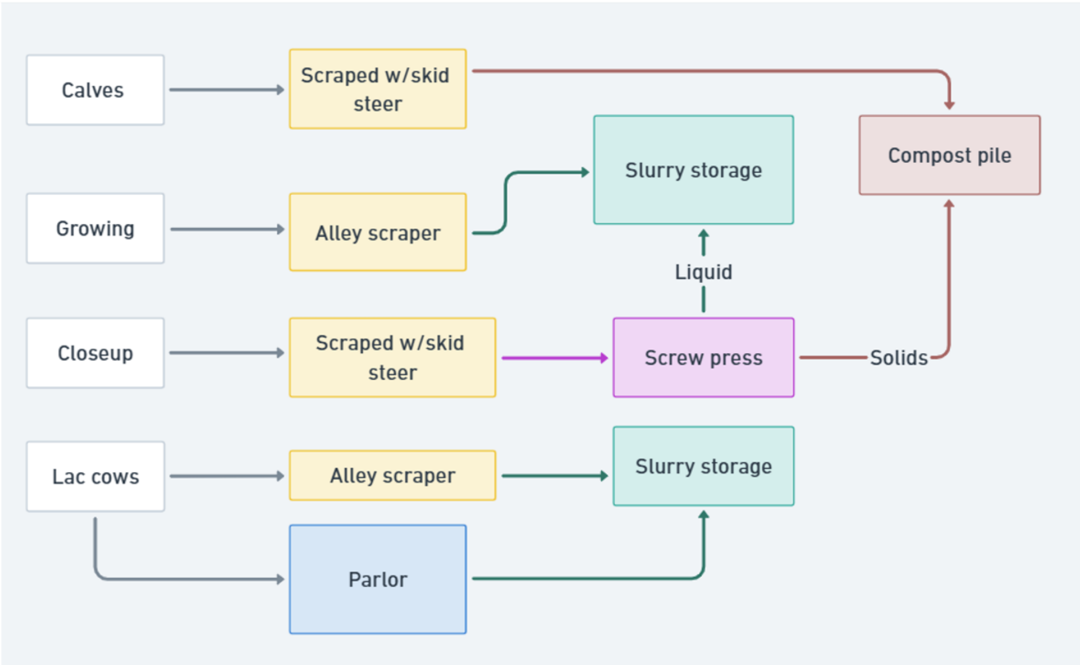

Figure 1: A real-world example of a simplistic manure management chain.

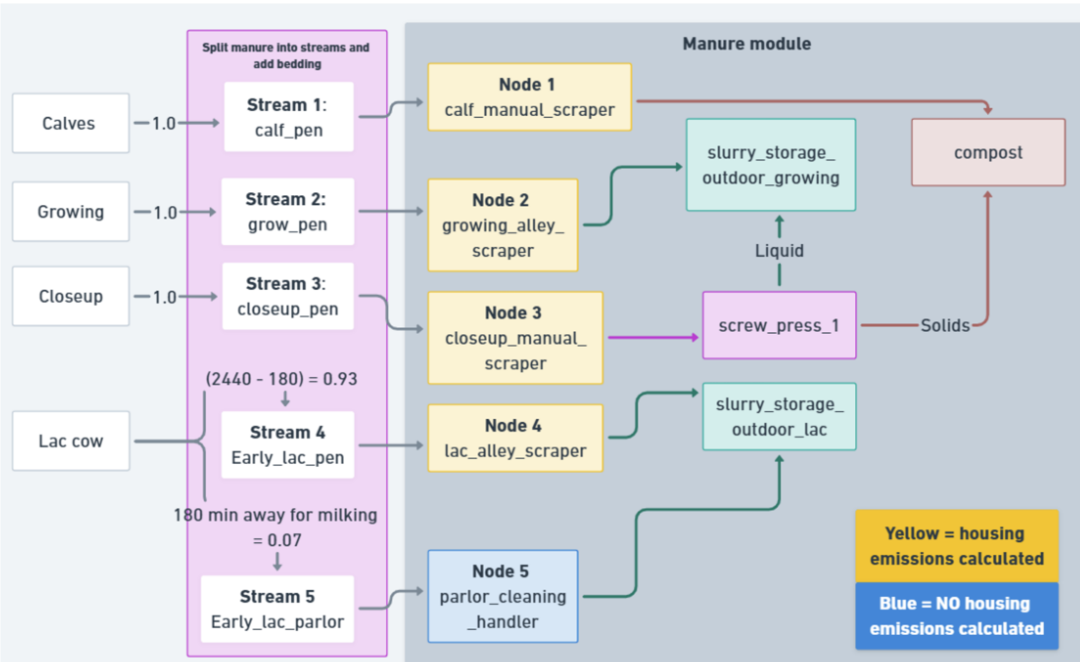

Figure 2: The manure management chain illustrated in the previous figure, with RuFaS-specific information reflected.

B. Overview of Required Inputs

There are two general categories of input required from the user:

1. Which processors are used and how do they work (processor-specific configurations) E.g., storage time length, use of a cover

2. Destination and proportion(s) allocated to destination(s) of manure passed from each processor

Information on inputs required and options available for each individual processor are outlined in the individual processor documentation.

For defining destinations and proportions, there are very few rules and restrictions on how manure can be moved between processors. The primary rule for defining processor destinations is that users cannot define loops, i.e., manure from a processor "downstream" in a manure management chain cannot return its manure to a processor "upstream".

The assignment of manure generated in the Animal Module (which also reflects the quantity of bedding used) to its destination in the Manure Module occurs in the Animal Module. See the Animal to Manure Connection section for more information.

C. General Assumptions of the Manure Module

1. All manure is recovered unless otherwise specifically stated. For example, all manure excreted by animals is assumed to be captured by manure handlers, all manure in a manure storage is assumed to be completely removed when the storage is emptied (unless explicitly noted in the specific processor's documentation), etc.

2. Manure moves through the manure management chain once, and only once, per day. With this, values are reported on a daily timestep, and represent the current state at the end of the given day (e.g. CH_4 emissions from this day, amount of N left in storage on this day, etc.).

   - Storage processors include a user-defined "storage time", during which manure accumulates in the storage until reaching the end of the time interval, at which point it is passed to the next processor or exported, if the storage comes last in the manure chain. The minimum storage time value is 1, thus manure cannot move through more than one storage per day.
   - For processors without a storage time option, manure is assumed to move through the processor immediately. E.g., on a single day, excreted manure may move through a handler, continuous mix digester, separator, and into composting storage, with each processor reporting the quantity and composition of manure they either passed to the next processor or are holding in storage.

3. The Manure Module adheres to the principle of mass balance, meaning, the quantity of manure nutrients remaining in manure is proportional to the quantity of the nutrient lost through biological or physical processes. The same is true for additions of mass/nutrients.

   - One current exception to this is N loss. N losses are not currently reflected in the total mass of manure; this exception will be addressed and corrected in the near future.

Manure Storage Assumptions

1. Received manure nutrients entering a storage are added to accumulated mass/nutrient quantities prior to calculating gas emissions each day.

2. When the end of the storage's user defined storage time is reached, the accumulated manure in storage is removed completely and entirely, and is passed to the next processor in the chain (e.g., a subsequent storage, field application, etc.).

3. Emptying intervals (in days) are defined by the user in equal lengths at this time, e.g., manure storage can be emptied every 3 months (e.g. in April, July, October, January) but cannot be emptied in April, then 2 months later in June, and then 3 months later in September. However, timing of storage emptying (e.g. April and October emptying vs. May and November emptying) can be set according to the general simulation dates to simulate more realistic manure removal behavior.

II. Animal and Manure Module Connection

A. Introduction

Though animals are typically fed and managed in groups, referred to as pens in RuFaS, manure from a single group of animals may not always remain in or be transported to the same location on the farm. For example, lactating cows who must travel to a milking parlor deposit some proportion of their manure outside of their pen. Or, in open lot systems, manure deposited in the concrete feed alley apron is often removed and managed separately from manure deposited on the lot surface. Additionally, more than one type of bedding may be used in a pen (e.g., no bedding during summer but straw provided in winter), which also eventually becomes part of manure. Considering this, RuFaS accommodates routing of manure from a single group (pen) of animals to multiple locations, which are then received by the Manure module. This document describes how a pen's total manure excretion and bedding are allocated to one or more discrete quantities of manure, referred to as manure streams, and passed to the Manure module. How the user configures this manure routing and bedding options and addition of bedding to manure are also described here.

Implementation in RuFaS

The Animal module is responsible for calculating manure excretion mass and composition based on animal attributes, ration composition and intake, and other factors. The Animal module must then pass this manure on to the Manure module, where manure nutrient gains and losses are quantified through field application or export. The Animal module currently calculates manure excretion at the pen level (i.e., values representative of the total manure from animals in a single pen), and several additional steps must occur before manure information can be sent to the Manure module. These steps include:

- Splitting manure from individual pens into one or more user-defined manure streams
- Repackaging manure information into the form and units utilized by the Manure module
- Applying bedding mass or nutrients to manure streams

The direct connection between Animal module, which captures the animal lifecycle, nutrient intake and excretion, and other factors, and the Manure module is a crucial part of the nutrient cycle in RuFaS. This document describes how the user defines the partitioning of manure from a single pen of animals into one or more manure streams, and the basic mathematical procedure for generating manure streams from excreted manure. Later it will review the options for and application of bedding to manure streams.

Classes

Table 2: List of Python classes for Animal to Manure Module connection.

Handlers	Description
Pen	pen.py; Translates pen-level manure information into the format and units required by the Manure module; Translates pen-level manure information into one or more ManureStream instance(s), if applicable, and creates accompanying PenManureData instances
PenManureData	animal_to_manure_connection.py; Generates PenManureData and ManureStream

Required User Inputs

Table 3: User inputs for Stream Formation and Bedding sections.

	Variable	Units	Description
Stream Formation	minutes_away_for_milking	minutes	The total time (minutes) per day that animals spend walking to/from the parlor, waiting in holding areas, or being actively milked. For non-lactating animals, this value may be omitted or entered as "null".
	first_parlor_processor	–	The name of the processor defined in the Manure module inputs that manure from the milking parlor should enter.
	parlor_stream_name	–	The user-defined name of this pen's parlor stream.
	stream_name	–	The user-defined name of this stream.
	bedding_name	–	The name of the specific bedding configuration used with this stream. See section 2 for more information.
	stream_proportion	0–1	The proportion of daily manure excreted outside of the parlor and while traveling to/from the parlor that enters this stream. All stream proportions within a pen must sum to 1.0. Optional if only one stream.
	first_processor	–	The name of the processor defined in the Manure module inputs that this ManureStream should enter.
Bedding	name	–	Unique identifier to reference this bedding configuration.
	bedding_type	–	The type of bedding material. Options: sand, straw, sawdust, manure solids, or none.
	bedding_mass_per_day	kg/day	Daily mass of fresh bedding added per animal per day (wet weight). Bedding mass is applied based on the total number of animals in the pen, and is not based on the stream proportion of the manure stream the bedding is assigned to.
	bedding_density	kg/m^3	Density of the bedding on a wet weight basis.
	bedding_dry_matter_content	fraction	Bedding dry matter content as a fraction of total mass.
	bedding_carbon_fraction	fraction	Bedding carbon content as a fraction of total mass.
	bedding_phosphorus_content	fraction	Bedding phosphorus content as a fraction of total mass.
	sand_removal_efficiency	fraction	Fraction of sand assumed to be removed immediately after manure removal.

Other inputs

- Instances of PenManure objects that represent manure excreted by single pens of animals in the Animal module. See Animal Excretions documentation for further details.
- Instances of Pen objects that represent pen information for single pens of animals in the Animal

module.

Expected Outputs

Stream Formation

- Instance(s) of a ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane potential, $m^3/kgVS$):
  - water
  - ammoniacal_nitrogen
  - nitrogen
  - phosphorus
  - potassium
  - ash
  - manure_degradable_volatile_solids
  - manure_non_degradable_volatile_solids
  - bedding_non_degradable_volatile_solids
  - total_solids
  - mass (equal to sum of water and total solids)
  - total volatile solids (equal to sum of degradable and non-degradable volatile solids)
  - volume
  - methane_production_potential
- A PenManureData class instance for each manure stream defined by the user, that represents the attributes of the animals and pen from which the manure originated. PenManureData instances include the following variables, reported by the Animal module:
  - num_animals: number of animals present in the pen from which the manure stream originated. Note that this variable does not correspond to the number of animals that generated the manure contained in the stream.
  - manure_deposition_surface_area: the surface area soiled with manure, used in the determination of housing emissions. See the Manure Handler document for information on determination of this value (MN.HDL.4).
  - animal_combination: the type of animal contained in the pen (calf, growing, closeup, or lactating).
  - pen_type: the type of pen (freestall, tiestall, open lot, or bedded pack barn) the manure in this stream originated from.
  - manure_urine_mass (kg): mass of urine in the manure.
  - manure_urine_nitrogen (kg): mass of urine N in the manure.
  - steam_type: the type of ManureStream (parlor stream or general stream).
  - first_processor: the name of the processor defined in the Manure module inputs that this ManureStream should enter.
  - total_bedding_mass (kg): the total mass of bedding added to this stream.
  - total_bedding_volume (m^3): the total volume of bedding added to this stream.

Bedding

- total_bedding_mass: The total mass of bedding added to the ManureStream each day (kg).
- total_bedding_volume: The total mass of bedding added to the ManureStream each day (kg).

B. Methodology

Stream Formation

How many discrete locations/quantities (i.e., manure streams) a single pen's manure is split into is determined by the user. There are two types of manure streams in the model:

Parlor: A stream representing manure deposited in or while traveling to/from the milking parlor, along with (fresh, non-recycled) wash water.

- A parlor stream is not directly defined by the user for each pen in the way general streams are; parlor streams are created automatically for each lactating pen. The user specifies the proportion of each lactating pen's total manure that enters the parlor stream via the minutes away for milking input. Time away from the pen for milking can vary greatly between farms but also is typically known by the farm management, as fetching and pushing cows to the parlor is an intentional daily activity. This makes time away from the pen for milking (minutes_away_for_milking) a simple and relatively reliable metric for determining the basic split of manure in the housing area vs. milking parlor.
- Only one parlor stream can be created per lactating cow pen. However, more than one milking parlor (destination for milking parlor manure) may be defined for a farm; the user simply assigns parlor streams to differing instances of a parlor cleaning handler, which represents the milking parlor. See Manure Handler documentation for more information.
- No bedding can be applied to parlor streams.

General: A stream representing all or a portion of non-parlor manure and bedding. Manure in this stream or streams represents manure excreted outside the milking parlor or traveling to/from the parlor.

- All pens must have at least one general stream defined.
- Theoretically, an infinite number of general streams can be defined per pen, as long as stream proportions are provided for all of them. However, most practical farm simulation scenarios will require only one or two general manure streams per pen.
- The number of general streams and the proportion of non-parlor manure entering each stream is directly defined by the user. Stream proportion logic is covered below.

General-type manure streams are defined in the animal inputs, within each Pen blob. Each pen includes a "manure_streams" array input in which the user defines the one or more general manure streams generated by this specific pen. The required inputs and their definitions are described above in the Inputs section; these inputs must be provided for each manure stream as outlined in the figure below.

```
        "manure_streams": [
            {
                "stream_name": "lac_exercise_lot",
                "bedding_name": "no_bedding",
                "stream_proportion": 0.25,
                "first_processor": "lac_alley_scraper"
            },
            {
                "stream_name": "lac_bedded_pen",
                "bedding_name": "lac_and_growing_sand",
                "stream_proportion": 0.75,
                "first_processor": "lac_bedded_pack"
            }
        ]
    }
```

Figure 3: If manure from a pen is routed to two different general manure streams, two sets of manure stream inputs must be defined. An example of a single pen's manure stream inputs is above - in this example, manure from this pen is routed to two locations, one representing an exercise lot, the other representing a bedded pack area.

Stream proportion logic

The stream proportion refers to the user-defined proportion (0 to 1) of manure generated by animals in a single pen that enters a specific manure stream. The translation of user-defined stream proportions into the effective stream proportion utilized by the model is described below. This effective proportion is referred to as the split ratio in the model code - it is the value by which total manure excretions are multiplied by to determine the quantity of any single nutrient allocated to a stream of any type (e.g., 20% (0.20) to parlor stream, 40% (0.40) to general stream 1, 40% (0.40) to general stream 2. The values are the split ratios).

Non-lactating pens If more than one general stream is defined for a single pen, a stream proportion must be provided for each stream, and manure will be allocated to streams according to these proportions. Essentially, the split ratio is equal to the user-defined stream proportion.

Lactating pens

Parlor streams: The user-defined minutes away for milking is used to calculate the stream proportion value for the milking parlor:

When:

$$parlor\_stream\_proportion = \frac{minutes\_away\_for\_milking}{1440}$$

[AN.PEN.1]

- minutes_away_for_milking: user-defined total minutes per day lactating cows spend out of the pen traveling to/from the parlor or being milked.
- 1440: minutes per day.

General streams Each general stream proportion represents the fraction of manure—excluding that deposited in or during travel to/from the milking parlor—that enters a specific manure stream. For example, if all manure outside the parlor enters one stream, its proportion would be 1.0. If manure outside the parlor was split evenly between two general streams, each stream's proportion would be

0.50. With this, the effective proportion of total manure entering this stream (the split ratio) must be calculated according to the parlor stream proportion.

First, we determine the proportion of manure entering any and all general streams vs. the parlor stream.

$$total\_general\_stream\_portion = 1 - parlor\_stream\_proportion$$

[AN.PEN.2]

When:

- parlor_stream_proportion: proportion of total manure allocated to the parlor stream, calculated in **[AN.PEN.1]**.

Once we have derived the proportion of manure entering any/all general streams, we multiply this proportion by the user-defined general stream proportion(s), to determine the split ratio for each general stream.

$$general\_stream\_proportion = total\_general\_stream\_proportion * stream\_proportion$$

[AN.PEN.3]

To continue with our example from above, if cows spent 200 minutes per day in the milking parlor, their split ratios would be as follows:

Parlor:

$$\frac{200}{1440} = 0.140$$

$$lac\_exercise\_lot : (1 - 0.139) * 0.25 = 0.215$$

$$lac\_bedded\_pen : (1 - 0.139) * 0.75 = 0.645$$

Recommended defaults

Although the time away from a pen is highly variable and specific to each farm's infrastructure and management, some recommended defaults for situations where actual information is not available are provided below.

Minutes away for milking

Recommended minutes_away_for_milking values for lactating cows are based on cow time budgets from published literature and are not official model defaults. The purpose of these time budgets at the time of writing is strictly related to estimating manure deposition in specific locations and is not related to larger Animal module activities. The use of time budgets for this purpose relies on the assumption that cows excrete urine and feces at a constant rate throughout the day and across various locations.

Open lot and bedded pack systems

The management of open lot and bedded pack barn pens often results in manure from a single pen being managed in multiple ways. These types of pens often have designated resting areas (i.e. the harrowed lot or bedded pack) where manure is managed in place for an extended period, as well as a separate feeding area, such as a concrete apron along a feed alley. Manure deposited in this area may be scraped back into the resting area, or it may be flushed, vacuumed or scraped to another separate storage area. Two distinct manure management routines within a single pen indicate that two manure streams should likely be used. If this is the case, the user may reference the tables below as a starting point for defining stream proportions representing the resting vs. feeding area. Defining two separate manure streams is recommended where appropriate to ensure that manure is not over-allocated to the parlor manure stream (where it does not contribute to housing emission estimations) and bedding is allocated appropriately.

Table 4: Recommended minutes_away_for_milking by pen type based on published literature. Remember, this does not necessarily reflect the model defaults.

Pen Type	minutes_away_for_milking	Proportion	Special Notes
Freestall[1]	200 m (3.33 h)	0.139	
Tiestall	200 m (3.33 h)	0.139	Limited data; value is based on the assumption that daily time away from pen for milking will be broadly similar between freestall and tiestall barns.[2]
Open lot	300 m (5 h)	0.208	Assumption that open lot pens tend to be considerably larger than indoor-housed systems (greater travel distance).[3]
Compost bedded pack barn	200 m (3.33 h)	0.139	Limited data; value is based on the assumption that daily time away from pen for milking will be broadly similar between freestall and tiestall barns.[2]

[1] **gomez2010time**, **espejo2007herd**

[2] **jewell2019prevalence**

[3] **meyer2019contract**

The following streams and proportions are based on literature data and are not official model defaults. They are provided for user informational purposes; they may be used as a starting point to adjust proportions specific to a farm, or can be used for theoretical modeling scenarios.

Table 5: Proportions of differnt streams for different animal classes.

Pen Type	Animal Class	Lactating	Closeup	Growing	Calves
Open lot or bedded pack barn [1]	General stream, lot	0.796	0.76	0.76	1
	General stream, feed alley	0.204	0.24	0.24	N/A
	minutes_away_for_milking	300	N/A	N/A	N/A
Bedded pack barn [2]	General stream, bedded pack	0.792	0.819	0.819	1
	General stream, feed alley	0.208	0.179	0.179	N/A
	minutes_away_for_milking	200	N/A	N/A	N/A

[1] Based on data from **meyer2019contract** and the assumption that open lot pens tend to be considerably larger than indoor-housed systems (greater travel distance).

[2] Based on data from **endres2007behavior**, which suggests behavior of cows in compost bedded pack barns is broadly similar to those in other housing systems. With this, freestall data from **gomez2010time** was used, which reported 4.3 hours of feeding time, 3.33 hours away for milking, and the remainder spent in the housing area.

Passing information to the Manure module

Once split ratios have been determined for each manure stream in each pen, Animal module manure and pen data can be translated into ManureStream objects. The variables contained in ManureStream are below; this dataclass is used both to move information from Animal to Manure module, as well as

move information between processors within the Manure module.
The majority of the variables are calculated in the Animal module and translated directly into a ManureStream object. Subbullets below a variable indicate how variables not directly reported by the Animal module are calculated.

Bedding

Bedding is assigned and applied by manure stream, not by pen. The following calculations described how bedding mass, nutrients, and volume are calculated, and how bedding is added to a manure stream.

Calculate total bedding mass

calc_total_bedding_mass

$$total\_bedding\_mass(kg) = bedding\_mass\_per\_day(kg/animal) * num\_animals * sand\_removal\_efficiency$$

[AN.PEN.7]

When:

- bedding_mass_per_day: The user-inputted bedding mass applied per animal per day. Note that this input is per total number of animals housed in the pen.
- num_animals: Animals in the pen from which the manure stream originated.
- sand_removal_efficiency: User-defined proportion of sand bedding removed via sand settling lane or other sand recovery technology. This value is only utilized if bedding type $\bar{s}$and.

Calculate total bedding volume

calc_total_bedding_volume

$$total\_bedding\_volume(m^3) = \frac{total\_bedding\_mass(kg)}{bedding\_density(kg/m^3)}$$

[AN.PEN.8]

When:

- total_bedding_mass (kg): the total mass of bedding applied to the manure stream each day, calculated in **[MN.PEN.7]**.
- bedding_density (kg/m^3): user-defined density of the specified bedding.

Calculate total bedding dry solids

calc_total_bedding_dry_solids

$$total\_bedding\_dry\_solids(kg) = total\_bedding\_mass(kg) * bedding\_dry\_matter\_content$$

[AN.PEN.9]

When:

- total_bedding_mass (kg): the total mass of bedding applied to the manure stream each day, calculated in **[MN.PEN.7]**.

- bedding_dry_matter_content: user-defined dry matter content of the specified bedding.

Calculate total bedding water

calc_total_bedding_water

$$total\_bedding\_water(kg) = total\_bedding\_mass(kg) * (1 - bedding\_dry\_matter\_content)$$

[AN.PEN.10]

When:

- total_bedding_mass (kg): the total mass of bedding applied to the manure stream each day, calculated in **[MN.PEN.7]**.
- bedding_dry_matter_content: user-defined dry matter content of the specified bedding.

Update ManureStream values to reflect bedding

_apply_bedding (pen.py)

Table 6: Calculated manure and bedding characteristics

Variable	Units	Calculation
water	kg	ManureStream water (kg) + bedding_water (kg)
ash	kg	if bedding type = sand, ash (kg) = total_bedding_mass (kg); otherwise, ash = 0
phosphorus	kg	ManureStream phosphorus + total_bedding_mass (kg) * bedding_phosphorus_content)
non_degradable _volatile_solids	kg	If bedding type = sand, ManureStream non_degradable_volatile_solids (kg); If bedding type is not sand, ManureStream non_degradable_volatile_solids (kg) + total_bedding_dry_solids (kg)
total_solids	kg	ManureStream total_solids (kg) + total_bedding_dry_solids (kg)
volume	m^3	ManureStream volume (m^3) + total_bedding_volume (m^3)

III. Manure Handler

A. Introduction

Manure handling refers to the method of managing manure in animal living spaces. For some housing/pen types, such as freestalls and tiestalls, this involves removing the manure from barn alleyways, and is usually accomplished using some form of scraping or flushing with water. Flush systems utilize water to remove the manure from alleyways, whereas scrape systems use little or no water. For other housing types like open lots and compost bedded pack barns, tillage or harrowing are commonly used to instead incorporate the manure, urine, bedding (if applicable) and air into the pack and allow it to dry.

The specific manure handler options in the RuFaS model are:

- **Flushing**

  Flushing involves using water to carry manure and soiled bedding away from housing areas. A series of pipes and channels direct water to wash the manure into a collection pit. Flushing helps maintain a cleaner environment for animals and reduces the buildup of manure on the floor. Flushing/recirculating may also be used to “lubricate” transport/pumping of manure from under-floor collection pits to the next manure management step. Flushing is commonly performed using recycled water from a solid liquid separator, liquid manure storage, or other liquid collection area, rather than using fresh water.

- **Manual Scraping**

  Scraping involves physically removing the manure and soiled bedding using equipment like tractor-mounted scrapers or skid steers, or the manual use of shovels/scrapers, etc. This method is typically used in barns with slatted or solid floors. Regular scraping prevents manure accumulation and provides a clean area for animals.

- **Alley Scraper**

  A scraper blade, typically pulled along by a chain or cable, moves slowly down a pen or alleyway on a set schedule several times a day, pushing or pulling manure out of the pen/alley. The function and outcome is similar to manual scraping, but alley scrapers are mechanized and typically operate automatically.

- **Parlor Cleaning**

  Parlor cleaning is a RuFaS-specific handler that represents cleaning of manure from milking units, parlor stall floors, and other areas in the milking parlor that require the use of fresh, non-recycled water. Parlor cleaning also can include an optional flush system capability, which functions in the same manner as the flush system described above and is used to wash manure from holding area floors.

Manure handlers in RuFaS have two primary functions: calculate and add cleaning water if used, and calculate indoor housing emissions of CH_4, CO_2, and NH_3 if applicable.

The following are important assumptions made in the manure handler submodules, as well as context on how manure handlers are used in the larger manure management chain:

1. All manure excreted and all bedding applied each day (if applicable) is assumed to be recovered by the manure handler.

2. Manure handlers in RuFaS are intended to be utilized, at this time, with manure deposited in housing areas that is cleaned frequently (e.g., removed daily or every few days), such as indoor barn floors, or concrete feed alleys in open lot or compost bedded pack pens. Handlers should not be used in manure streams where manure resides for a long period of time, such as streams representing an open lot or bedded pack manure pack, that have a designated processor (i.e., Open Lot or Bedded Pack). Users can consider the manure handler functionality to be “built in” to these processors. Defining a handler prior to one of these types of processors will result in double-counting of housing emissions.

3. If a manure handler is used, it must be defined first in the manure management chain. This prevents inappropriate calculation of housing emissions “downstream” when manure is no longer in the animal housing area. Handlers are meant to represent manure handling in animal living spaces; other manure handling/transport systems such as pumps, augers, belts, etc. are not explicitly modeled in RuFaS.

Classes

Table 7: The classes of handlers that are used in this module.

Handlers	Description
Handler (Processor)	handler.py contanins basic methods that are applicable to all handlers
Single StreamHandler (Handler)	single_stream_handler.py contains alley screaper, manual scraping, and flush system-specific methods
ParlorCleaning (Handler)	parlor_cleaning.py contains parlor cleaning-specific methods

Required User Inputs

Table 8: Required inputs for the Manure Handler (refreshed_manure_management.json)

Variable	Units	Description
Name	none	Unique identifier of the specific handler configuration used.
Type	none	The type (alley scraper, manual scraping, flushing, or parlor scraping) of manure handler.
cleaning_water_use_-amount	L/animal/day	The rate of water use (L per animal per day, total of fresh and/or recycled water) by a specified manure handler.
cleaning_water_recycle_-fraction	none	The fraction (0.XX) of cleaning water used by a specified handler that is from recycled (not fresh) water sources.
use_parlor_flush	true/false	True/false indication for if a parlor flush (which may be partially or fully recycled water) is used in addition to routine parlor water cleaning with fresh water.

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen
- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids

- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids
- total_solids
- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)
- volume
- methane_production_potential

Manure handlers also make use of information contained in the PenManureData subclass. This includes the following information specific to the pen from which the manure was generated:

- num_animals: number of animals in the pen
- animal_combination: type of animals in the pen (calves, growing post-weaning heifers through closeup period), closeup (cows and heifers within user-defined closeup period), or lactating cows)
- Manure deposition surface area (m2): surface area soiled with manure. See section below on determining soiled surface area
- pen_type: the type of pen the manure was generated in (freestall, tiestall, open lot, or bedded pack barn)
- daily_urine (kg): mass of urine in the manure
- urine_nitrogen (kg): mass of urine N in the manure
- stream_type: the type of manure stream the manure is in (parlor or general stream)

Note that stream_type is not a user input, it is automatically assigned. Parlor streams are created automatically and all other streams are assigned “general”.

Expected Outputs

- All handler types:
  - ManureStream variables
  - total_cleaning_water_volume (m^3): total volume of fresh (non-recycled) cleaning water used for, and ultimately added to, a single manure stream on a single simulation day by the manure handler
  - barn_temperature ($\circ C$): ambient temperature in barn
- Alley scraper, manual scraping, flush system:
  - housing_methane (kg): daily CH_4 (kg) emitted from manure on barn floors
  - housing_ammonia_N_emissions: (kg): daily NH_3-N (kg) emitted from manure on barn floors
  - housing_carbon_dioxide (kg): daily CO_2 (kg) emitted from manure on barn floors

B. Methodology

Calculate barn temperature

_determine_barn_temperature
Air temperature inside barns is, in general, slightly cooler than the outdoors during the day, and slightly-to-considerably warmer than the outdoor temperature overnight, depending on the season. With this, basic bounds were implemented in the determination of barn temperature. These bounds were suggested by manure SMEs and are supported by barn temperature ranges reported in Bucklin et al., 2009 (Florida, upper limit). The lower bound (5°C) suggested by SMEs was based on general industry standards/conditions.

If ambient (outdoor) air temperature < 5°C, barn temperature = 5°C
If ambient (outdoor) air temperature > 5° & < 30°C, barn temperature = air temperature
If ambient (outdoor) air temperature > 30°C, barn temperature = 30°C

Calculate cleaning/flushing fresh water addition

determine_handler_cleaning_water_volume
The following method calculates the volume of fresh (non-recycled) cleaning water used for, and ultimately added to, a single manure stream on a single simulation day by the manure handler. This method is used in all handler types; however, for parlor cleaning handlers, this method is only utilized if "use_parlor_flush" is entered as true. If no water is used with the handler (user inputted cleaning water use rate = 0), the function will return 0.

$$cleaning\_water\ (L) = num\_animals * (cleaning\_water\_use\_rate * (1 - cleaning\_water\_recycle\_fraction))$$

[MN.HDL.1]

When:

- num_animals: the number of animals in the pen assigned to the handler
- cleaning_water_use_rate (L/animal/d): the user-provided cleaning water use rate for the handler
- cleaning_water_recycle_fraction: the proportion of cleaning water that is from recycled, non-fresh sources

Total fresh water added to manure (L) each day is calculated as follows:

$$total\_cleaning\_water(m^3) = (cleaning\_water(L) + fresh\_water\_volume\_used\_for\_milking(L)) * LITERS\_TO\_CUBIC\_METERS$$

[MN.HDL.2]

When:

- LITERS_TO_CUBIC_METERS: a constant set to 0.997 kg/m^3

Calculate milking parlor fresh water addition

The following method calculates the volume of fresh water used for, and ultimately added to, a single manure stream on a single simulation day by the manure handler for parlor cleaning/sanitation purposes. This calculation occurs only in the parlor_cleaning handler and water added is considered to be 100% fresh, non-recycled water, as fresh water is required for cleaning of milking units, floors, etc. The following function is implicit (occurs automatically with any manure stream assigned to the parlor cleaning handler type) and is based at this time on a default value fresh water use rate.

$$fresh\_water\_volume\_used\_for\_milking\ (L) = num\_animals * MILKING\_FRESH\_WATER\_USE\_RATE$$

[MN.HDL.3]

When:

- num_animals: the number of animals in the pen assigned to the handler
- MILKING_FRESH_WATER_USE_RATE (30.0 L/animal/d): the default fresh water use rate for cleaning equipment/floors in the milking parlor

Calculate housing emissions

The calculation of housing emissions according to the methods described below is applicable only to areas where manure resides for short periods of time, i.e., tiestall and freestall housing systems, or, in open lot or bedded pack pens, manure deposited in a feed alley, exercise lot, etc. Housing emissions from bedded packs and open lot manure packs, where manure accumulates over weeks or months, are determined according to alternative methods due to the longer-term residence of manure in these pen types; see respective bedded pack barn and open lot documentation files for further info.

The methodology used to calculated housing emissions is adapted from the IFSM (Rotz, Corson, et al., 2023). Of note is that the equations described here were determined based on emissions in tiestall and freestall housing systems. In RuFaS, these equations may also be applied to determine emissions from manure deposited in other concrete areas and retained for a short time, such as open lot or bedded pack feed alleys. These systems were not represented in the original datasets used to generate these equations, and results should be interpreted with caution; however, the portion of manure deposited in feed alleys or other areas is relatively small, and emissions from this area are expected to be a small contributor to total GHG and nutrient losses.

Additionally, the following calculations are only utilized by the manual scraping, flush system, and alley scraper handler types. For the parlor cleaning handler type, manure in a parlor is assumed to be cleaned from floors/surfaces frequently and thus housing emissions should not be estimated.

Calculate housing CH_4 emissions

Manure deposited on barn floors or other housing areas is a small source of CH_4 emissions. The following method, from Chianese, Rotz, and Richard, 2009, is an empirically-derived equation consisting of a simple relationship between manure CH_4 emission rate, ambient temperature in the barn, and surface area covered by manure.

> The soiled surface area (m2) value is calculated in the Animal module, using pen-level information (number of stalls, number of cows, etc.) and the soiled surface area value is passed to the manure module directly. As no housing emissions are considered to be emitted in the milking parlor, the total soiled surface area for each pen is apportioned to the pen's general manure streams(s) based on the user-provided stream proportion value(s).

The total surface area soiled with manure, based on the total number of animals, is as follows:

$$soiled\_surface\_area\ (m^2) = num\_stalls * surface\_area\ (m^2/animal)$$

[MN.HDL.4]

When:

- num_stalls: the number of stalls in the pen assigned to the handler

Table 9: The surface area per stall that is covered with manure is a function of the pen type and type of animal housed.

System Type	Lactating Cows	Closeup, Growing, or Calves
Tiestall	1.2 m^2	1.0 m^2
Freestall	3.5 m^2	2.5 m^2
Bedded Pack	5.0 m^2	5.0 m^2
Open Lot	3.0 m^2	3.0 m^2

- surface_area (m^2/animal): surface area per stall, provided in Table 8.

We use this surface area value to determine CH_4 emissions (kg) each day

$$housing\ CH_4\ (kg) = max^1(0.0, 0.13 * barn\_temperature(C)) * \frac{soiled\_surface\_area\ (m^2)}{1000}$$

[MN.MET.1]

When:

- barn_temperature ($°C$): ambient air temperature in the barn. Note that the max notation ensures that the temperature scaling parameter cannot be less than 0.13.
- soiled_surface_area (m^2): the total surface area soiled with manure

Calculate housing CO_2 emissions Housing CO_2 emissions are estimated similarly to CH_4 emissions, using an empirical equation based on soiled surface area and barn air temperature.

$$housing\ CO_2\ (kg) = max^1(0.0, 0.0065 + barn\_temperature(C)) * soiled\_surface\_area\ (m^2)$$

[MN.HDL.5]

When:

- barn_temperature ($°C$): ambient air temperature in the barn. Note that the max notation ensures that the temperature scaling parameter cannot be less than 0.13.
- soiled_surface_area (m^2): the total surface area soiled with manure

Calculate volatile solids losses The total quantity of VS loss through microbial degradation is estimated based on CH_4 emissions.

$$total\_VS\_loss\ (kg) = housing\ CH_4\ (kg) * VS\_TO\_METHANE\_LOSS\_RATIO$$

[MN.HDL.6]

When:

- housing CH_4 (kg): daily emissions of CH_4 from manure on barn floors, calculated in **[MN.MET.1]**.

- VS_TO_METHANE_LOSS_RATIO (kg): a constant set to 6.665 representing the kg of total VS degraded per kg of CH_4 emitted from manure. See the Slurry Storage section for more information on derivation of this constant.

The fraction of volatile solids lost from the manure degradable (VSd) and non-degradable VS (VSnd) fractions is assumed to be equal to the proportion of these fractions in manure entering the processor. No VS losses are attributed to bedding VSnd in manure handlers.

$$degradable\_VS\_frac = (\frac{degradable\_VS}{total\_VS})$$

[MN.ADG.7]

Calculate housing NH_3 emissions Ammonia N loss from manure is predicted using methods based on Rotz and Oenema, 2006. Here, estimated NH_3-N volatilized from the surface of manure is based on a function where the NH_3-N is transported to the free atmosphere through a pathway with a finite resistance. Daily NH_3-N loss depends primarily on pen type (which determines soiled surface area), manure TAN content, and barn air temperature. The same base equation for NH_3-N volatilization is utilized for estimation of both housing and storage NH_3-N emissions.

In the original work of Rotz and Oenema, 2006, housing NH_3-N emissions are based only on urine (not manure) TAN concentration, rather than total manure (urine + feces) TAN. Within RuFaS, urine nutrients are not tracked separately from feces/manure. Therefore, the model assumes, for simplicity, that all TAN originates from urine and thus animal-excreted urine TAN = manure TAN. However, to maintain consistency with documentation, what would ordinarily in RuFaS be referred to as manure TAN within RuFaS is referred to here as urine TAN, though urine mass here (kg/d) is reflective of actual estimated urine mass, not manure (urine + feces) mass.

- First, we need to derive the value of the equilibrium coefficient Q for the NH_3 gas in the air for a given concentration of TAN in stored slurry using Henry's law of distribution.

The concentration of NH_3 in the free atmosphere is assumed to be zero. Since Q is a function of the Henry's law coefficient K_h and a dissociation of ammonium coefficient K_a, we will calculate those first.

Henry's law coefficient (K_h)

$$K_h = 10^{\frac{1478}{barn\ temperature - 1.69}}$$

[MN.AMM.1]

When:

- Barn temperature (K): ambient air temperature in the barn

Dissociation coefficient of ammonium (K_a)

$$K_a = 1 + 10^{(0.09018 + (\frac{2729.9}{barn\ temperature} - pH)}$$

[MN.AMM.2]

When:

- Barn temperature (K): ambient air temperature in the barn
- pH: the pH of the manure solution, set to 7.5 by default

Equilibrium coefficient (Q)

$$Q = K_h * K_a$$

[MN.AMM.3]

When:

- K_h: Henry's law coefficient, calculated in **[MN.AMM.1]**.
- K_a: Dissociation coefficient of ammonium, calculated in **[MN.AMM.2]**.

- Second, we determine the effective total resistance value of NH_3 transport from manure solution to the surface and from the manure surface to the atmosphere.

$$ammonia\_resistance = HOUSING\_SPECIFIC\_CONSTANT * (1 - 0.027 * (20.0 - barn\ temperature))$$

[MN.AMM.4]

When:

  - HOUSING_SPECIFIC_CONSTANT: a housing-specific constant value, set to 260 s/m.
  - Barn temperature ($°C$): ambient air temperature in the barn.

- Next, the rate of NH_3-N loss in kg N/m^2 from manure on the barn floor is calculated

$$NH_3N\ emission\ rate\ (kg\ N/m^2) = \frac{(TAN*c*y)}{(r*M*Q)}$$

[MN.AMM.5]

When:

  - TAN (kg N/m^2): Mass of ammoniacal N in barn floor manure
  - c: time conversion constant (86400 s per d)
  - y: manure density, set to 990 kg/m^3
  - r: the resistance of NH_3 transfer from solution to manure surface, and from manure surface to atmosphere, determined in **[MN.AMM.4]**
  - M: urine mass (received from animal module)
  - Q: equilibrium coefficient, calculated in **[MN.AMM.3]**

- Lastly, we calculate total NH_3-N emissions (kg), based on the emission rate we just calculated and the soiled surface area.

$$NH_3N\ emissions\ (kg) = NH_3N\_rate * soiled\_surface\_area$$

[MN.AMM.7]

When:

- NH_3N_rate (kg N/m^2): Rate of NH_3-N loss (kg/m^2) from manure, calculated in **[MN.AMM.5]**
- soiled_surface_area (m^2): the total surface area soiled with manure

C. Manure composition update

Below is a summary of updates to ManureStream variables.

> The formulas below may be a summarization of multiple steps detailed above, and are intended to provide an overview of what mass losses/gains are reflected in the value of each variable.

Equations in the table below are in the format of “Output/exiting value” = “Entering processor value” +/- XYZ. “Entering” refers to the manure entering the processor (i.e., being loaded into the digester) and output variables reflect the manure leaving the processor (i.e., digestate leaving the digester). The Calculation column describes how the variable is updated in the processor.

Table 10: Calculated manure storage variables and their units

Variable	Units	Calculation
water	kg	Entering water (kg) + (total_cleaning_water (m^3) × WATER_DENSITY_KG_PER_M3)
total_ammoniacal_nitrogen	kg	min(0, entering ammoniacal_nitrogen (kg) − NH_3-N emissions (kg))
nitrogen	kg	min(0, entering nitrogen (kg) − NH_3-N emissions (kg))
phosphorus	kg	Entering phosphorus (kg)
potassium	kg	Entering potassium (kg)
ash	kg	Entering ash (kg)
non_degradable_volatile_-solids		Entering non_degradable_volatile_solids − (total_VS_loss × (1 − degradable_VS_frac))
degradable_volatile_solids		Entering degradable_volatile_solids − (total_VS_loss × degradable_VS_frac)
total_solids	kg	Entering total_solids − total_VS_loss
volume	m^3	$\frac{\text{Entering volume} - \text{total\_VS\_loss}}{\text{ManureConstants.SLURRY\_MANURE\_DENSITY}}$ [1,2]

[1] WATER_DENSITY_KG_PER_M3: a constant with a value of 0.997 kg/m^3.

[2] Note that for parlor cleaning handlers, the value of all housing emissions variables will be reported as 0.

IV. Solid Liquid Separators

A. Introduction

A solid-liquid separator (SLS) is a specialized piece of equipment or system designed to separate larger, solid particles from the liquid component of manure. This type of system may be implemented for a variety of reasons, such as to improve ease of handling of manure liquid, improve the nutrient concentration in manure liquid, reduce storage volume required for manure lagoons or other storage systems, prevent solid accumulation in a covered manure storage, or reclaim manure solids for use as bedding or compost.

The solid fraction of manure slurry is composed of fibers originating primarily from manure, but also from feed, bedding, and other environmental sources. Once mechanically separated, the moisture content of the solid fraction ranges from 70% to 90%, depending on the specific separator system used. To further reduce moisture content of the separated solids, an additional dewatering or drying step may be incorporated. Manure solids may be directly used or sold for animal bedding (after stabilization), composted, or land applied.

After the manure solids are mechanically separated, the remaining liquid fraction is composed of small particles, water, and most of the nutrients present in the manure. Because the bulk of relevant nutrients (e.g., P, N, K) are mainly associated with the small particles which remain with the liquid fraction after separation occurs, separating the larger, less nutrient-dense particles out from the slurry liquid enriches the concentration of these nutrients in the liquid fraction. Solid-liquid separation also makes the manure slurry easier to pump, transport, and apply to fields, as large particles that may clog lines and sprayers are removed.

Implementation in RuFaS

In RuFaS, solid liquid separators currently represent mechanical, short retention time pieces of equipment. GHG emissions and other nutrient transformations from long-retention separators such as weeping walls or settling basins are not yet represented in the model. Because of this, the impact of these longer retention separators can only be approximated via modification of the separation efficiencies of the short retention time methods.

Currently, SLS inputs are simply nutrient-specific separation efficiencies that reflect the proportion of a specific nutrient that is separated into the solids fraction. With this, essentially any type of mechanical manure separation system can be modeled, as long as effective separation efficiencies are known. Manure is assumed to be loaded, processed, and passed to the next processor within a single day. Default separation efficiencies exist for two common types of SLS - a screw press and a rotary screen. The separated manure solids and remaining liquid fraction are quantified and can be managed in separate, downstream manure management systems in RuFaS. However, separated solids cannot be directly recycled (i.e., utilized with the specific nutrient composition of those separated solids) "upstream" as bedding.

Classes

Table 11: List of Python classes for solid liquid separators.

Handlers	**Description**
Separator (Processor)	separator.py

Required User Inputs

Table 12: The required inputs of this section.

Variable	**Definition (units)**	**Description**
Name	none	Unique identifier of the specific separator configuration used.

Variable	Definition (units)	Description
type	rotary screen or screw press	The type of solid liquid separator ; selection of this input indicates which default separations efficiencies should be used, however, any and all separation efficiencies may also be overwritten by user input.
separated_solids_dry_matter	none	percent dry matter by mass of the manure solids post solid-liquid separation.
total_solids_efficiency	none	the proportion of manure total solids (TS) that are separated into the manure solids fraction by the separator.
volatile_solids_efficiency	none	the proportion of manure volatile solids (VS) that are separated into the manure solids fraction by the separator.
nitrogen_efficiency	none	the proportion of manure total nitrogen that is separated into the manure solids fraction by the separator.
ammoniacal_nitrogen_efficiency	none	the proportion of manure ammoniacal nitrogen that is separated into the manure solids fraction by the separator.
phosphorus_efficiency	none	the proportion of manure phosphorus that is separated into the manure solids fraction by the separator.
potassium_efficiency	none	the proportion of manure potassium that is separated into the manure solids fraction by the separator.
ash_efficiency	none	the proportion of manure ash that is separated into the manure solids fraction by the separator.

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen
- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids
- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids

- total_solids
- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)
- volume
- methane_production_potential

Expected Outputs

- Two sets of ManureStream variables, representing the separated solids ("SeparatedSolids") and liquid fraction ("SeparatedLiquid"), respectively

B. Methodology

Separate nutrients

The equations below describe the general process for separating nutrients between liquid and solid fractions. The general equation for nutrient removal is as follows:

$$Separated\ solids\ nutrient\ content = received\ manure\ nutrient * separation\ efficiency$$

[MN.SEP.1]

$$Separated\ liquid\ nutrient\ content = received\ manure\ nutrient * (1 - separationefficiency)$$

[MN.SEP.2]

When:

- Received manure nutrient (kg): quantity of the specific nutrient in manure being loaded into the separator on a single day.

Nutrients that are separated according to this pattern are presented in the table below. The default separation efficiencies, as well as minimum and maximum allowable separation efficiencies, for the rotary screen and screw press separators are also provided (Hegg, R. E. Larson, and Moore, 1981; Mukhtar, Sweeten, and Auvermann, 1999; Varma et al., 2021). Note that separate removal efficiencies for manure degradable and non-degradable VS or bedding non-degradable VS are not specifiable at this time; all VS fractions are assumed to be removed at the rate of total VS removal.

Table 13: The default, minimums, and maximums for various separation efficiencies.

Type	Variable	Default	Min	Max
Rotary Screen	Total solids	0.35	0.25	0.4
	Nitrogen	0.3	0.25	0.35
	Ammoniacal N	0.15	0.1	0.2
	Phosphorus	0.4	0.3	0.45
	Potassium	0.15	0.05	0.2
	Ash	0.2	0.05	0.3
	Volatile solids	0.35	0.3	0.45
Screw Press	Total solids	0.25	0.15	0.35
	Nitrogen	0.3	0.2	0.35
	Ammoniacal N	0.15	0.05	0.2
	Phosphorus	0.2	0.1	0.35
	Potassium	0.23	0.15	0.35
	Ash	0.2	0.05	0.3
	Volatile solids	0.25	0.2	0.4

[1] (Jørgensen and Jensen, 2009; Fournel et al., 2019)

Calculate total mass and water

Separated solids fraction Total mass of the separated solids fraction is determined by dividing the mass of separated solids by the user-inputted separated solids dry matter content.

$$Separated\ solids\ mass\ (kg) = \frac{Separated\ solids\ TS\ (kg)}{Separated\ solids\ \%DM}$$

[MN.SEP.3]

When:

- Separated solids TS (kg): the mass of solids in the separated manure solids fraction
- Separated solids %DM: user-inputted separated solids dry matter content

The mass of water in the separated solids fraction can then be determined as follows, given the assumption that total mass is equal to the sum of solids plus water:

$$Separated\ solids\ water\ (kg) = Separated\ solids\ mass\ (kg) - Separated\ solids\ TS\ (kg)$$

Separated liquid fraction

The total mass of the separated liquid fraction, as well as the mass of water, are simply equal to the mass and water in manure loaded into the separator, minus the quantity of each respective value partitioned into the separated solids fraction.

$$Separated\ solids\ mass\ (kg) = \\ Received\ mass\ (kg) - Separated\ solids\ mass\ (kg) Separated\ solids\ water\ (kg) = \\ Received\ water\ (kg) - Separated\ solids\ water\ (kg)$$

Calculate volume

Volume of each separated fraction is determined by dividing the mass of each fraction by its respective density:

$$Separated\ solids\ volume\ m^3 = \frac{Separated\ solids\ mass}{SOLID\_MANURE\_DENSITY}$$

[MN.SEP.4]

$$Separated\ liquid\ volume\ m^3 = \frac{Separated\ liquid\ mass}{LIQUID\_MANURE\_DENSITY}$$

[MN.SEP.5]

When:

- SOLID_MANURE_DENSITY (kg/ m^3): The default density of solid manure, set to 700 kg/ m^3
- LIQUID_MANURE_DENSITY(kg/ m^3): The default density of liquid manure, set to 900 kg/ m^3

C. Manure composition update

Below is a summary of updates to ManureStream variables. Note that the formulas below may be a summarization of multiple steps detailed above, and are intended to provide an overview of what mass losses/gains are reflected in the value of each variable.
Equations in the table below are in the format of "Output/exiting value" = "Entering processor value" +/- XYZ. "Entering" refers to the manure entering the processor (i.e., being loaded into the digester) and output variables reflect the manure leaving the processor (i.e., digestate leaving the digester). The Calculation column describes how the variable is updated in the processor.

Table 14: Solids and liquid fraction calculations for separated manure components

Variable	Units	Calculation (Solids Fraction)	Calculation (Liquid Fraction)
water	kg	Separated solids mass (kg) − Separated solids TS (kg)	Entering water (kg) − Separated solids water (kg)
total_ammoniacal _-nitrogen	kg	Entering ammoniacal nitrogen (kg) × TAN removal efficiency	Entering ammoniacal nitrogen (kg) × (1 − TAN removal efficiency)
nitrogen	kg	Entering nitrogen (kg) × N removal efficiency	Entering nitrogen (kg) × (1 − N removal efficiency)
phosphorus	kg	Entering phosphorus (kg) × phosphorus removal efficiency	Entering phosphorus (kg) × (1 − phosphorus removal efficiency)
potassium	kg	Entering potassium (kg) × potassium removal efficiency	Entering potassium (kg) × (1 − potassium removal efficiency)
ash	kg	Entering ash (kg) × potassium removal efficiency	Entering ash (kg) × (1 − potassium removal efficiency)
manure_degradable _volatile_solids	-	Entering manure VSd (kg) × VS removal efficiency × degradable_VS_frac	Entering VSd (kg) × (1 − VS removal efficiency) × degradable_VS_frac

Continued on next page

Variable	Units	Calculation (Solids Fraction)	Calculation (Liquid Fraction)
manure_non_degrad-able _volatile_solids	-	Entering manure VSnd (kg) $\times$ VS removal efficiency $\times$ (1 $-$ degradable_VS_frac)	Entering VSnd (kg) $\times$ (1 $-$ VS removal efficiency) $\times$ (1 $-$ degradable_VS_frac)
bedding_non_degrad-able _volatile_solids	-	Entering manure VSnd (kg) $\times$ VS removal efficiency $\times$ (1 $-$ degradable_VS_frac)	Entering VSnd (kg) $\times$ (1 $-$ VS removal efficiency) $\times$ (1 $-$ degradable_VS_frac)
total_solids	kg	Entering TS (kg) $\times$ TS removal efficiency	Entering TS (kg) $\times$ (1 $-$ TS removal efficiency)
volume	m^3	$\frac{\text{Separated solids mass (kg)}}{\text{SOLID\_MANURE\_DENSITY}}$	$\frac{\text{Separated liquid mass (kg)}}{\text{LIQUID\_MANURE\_DENSITY}}$

V. Anaerobic Digestion

A. Introduction

Anaerobic digestion is a process where dairy cow manure is treated in an oxygen-free (anaerobic) environment to produce biogas, containing approximately 60% CH_4, 40% CO_2, which can be utilized as an on-farm or exported energy source. Additional benefits of anaerobic digestion include reduced manure odor, improved stability and quality of manure for fertilization purposes, and reductions in nutrient losses via undesirable greenhouse gas emissions.
Manure can undergo anaerobic digestion in a specialized anaerobic digestion chamber/system, usually referred to simply as a 'digester', or can occur in other manure storage systems that create anaerobic conditions, such as anaerobic lagoons.

Implementation of RuFaS

The anaerobic digestion submodule in RuFaS currently represents anaerobic digestion within an enclosed, mechanical digester system. The following assumptions are made in the RuFaS anaerobic digestion submodule:

- The digester represented is assumed to be a mesophilic, continuous stirred-tank reactor (CSTR).
- Manure is loaded into the digester and removed from the digester once daily.
- The digester is fully functional at the start of the simulation, i.e., the digester is full of substrate and the microbial population has stabilized.
- Residence time of manure in the digester is not modeled. As a result, the composition of effluent exiting the digester each day is identical to the influent composition, with the exception of nutrient losses or transformations that occur in the digester.
- At this time, CH_4 generation and volatile solids destruction is based strictly on the quantity of manure volatile solids loaded; other factors are not considered at this time.

The anaerobic submodule has two primary functions in RuFaS:

- Estimate daily methane production (kg/d) based on the daily mass of manure volatile solids (VS) loaded into the digester.
  - VS loading is dependent on the number and type of animals contributing manure to the digester, the diet of the animals, quantity and type of bedding, and any upstream manure handling processes, e.g., solid liquid separation, water addition, etc.
- Update the composition of the liquid manure effluent leaving the digester, which enters the anaerobic lagoon.
  - Reflecting VS loss in effluent exiting the digester is essential to capture the reduction in methane emissions from digestate compared to undigested, liquid manure, as well as changes in proportion of inorganic (ammoniacal) to total nitrogen.

Classes

Table 15: List of Python classes for anaerobic digestors.

Digester	Description
AnaerobicDigester	anaerobic_digester.py inherits behavior from the base class, Digester; however, at this time, there is no functionality included in the base Digester class. Therefore all methods are contained within the anaerobic_digestion.py child class.

Required User Inputs

Table 16: Required inputs for this section.

Variable	Units	Description
name	none	Unique identifier of the specific anaerobic digester configuration used.
hydraulic_retention_time	days	Number of days manure spends in the anaerobic digester. Note that this variable is not utilized directly by the module, but is utilized by the Economics, Emissions and Energy module.
anaerobic_digestion _temperature_set_point	°C	Temperature set point for the anaerobic digestion. This input is utilized in the conversion of CH_4 and CO_2 generation volume to mass; it does not directly influence CH_4 emissions.
biogas_leakage_fraction	none	Fraction of biogas generated in the anaerobic digester that escapes to the atmosphere through unintended leakage and is not collected by the gas capture system .

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane production potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen
- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids
- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids

- total_solids
- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)
- volume
- methane_production_potential

Expected Outputs

- captured_biogas_volume (m^3): Captured biogas (assumed to be composed of 40% CO_2, 60% CH_4) volume after accounting for leakage on the current day
- captured_methane_volume (m^3): Captured methane volume on the current day, after accounting for leakage
- methane_leakage_mass (kg): Mass of CH_4 lost to the atmosphere through unintended leakage on the current day. This variable is expressed as mass as opposed to volume as CH_4 emissions are reported in kg in the rest of the manure module and other RuFaS modules

B. Methodology

Calculate Daily Methane Generation

Description: calculates volume of methane (CH_4) generated from a CSTR digester. Methane generation is estimated from the daily loading of manure volatile solids (VS). Degradable (VSd) and non-degradable (VSnd) VS are tracked separately but the ratio of VSd : VSnd does not affect CH_4 estimation; only the total quantity of VS is considered. To perform this calculation, we make several assumptions/simplifications:

- The ratio of chemical oxygen demand (COD): VS in dairy manure is assumed to be 1.2 to 1. 1 kg of COD can generate 0.4 kg of methane. Therefore, 1 kg VS reduction/degradation in the anaerobic digester = 1.2 kg COD = 480 L CH_4. In other words, each kg VS of reduced is assumed to generate 480 L of CH_4.
- A CSTR reduces manure VS content by approximately 50%. This is a generalization across CSTR digesters of varying efficiencies, based on expert opinion from W. Liao (MSU) and A. Leytem (USDA-ARS).
- Considering that 480 L of CH_4 are generated per kg of VS destroyed, and approximately 50% of VS loaded are anticipated to be destroyed, we estimate CH_4 generation by assuming 240 L CH_4 are generated per VS kg loaded into the digester; this is also in alignment with the IPCC (2019) Tier II manure CH_4 Bo value.
- Lastly, the volumetric ratio of CH_4 to CO_2 generation in the CSTR is assumed to be 6:4 (e.g. generation of 60% CH_4, 40% CO_2 biogas) based on commonly cited digester performance metrics (e.g., EPA, Fernández et al., 2015). The total quantity of VS destroyed in anaerobic digestion is then assumed to be equal to the quantity of CH_4 and CO_2 generated in the digester. This assumption is made due to a lack of data on destruction of degradable vs. non-degradable VS in anaerobic digestion.

$$generated\_methane\_volume\ (m^3) = \\ ACHIEVABLE\_METHANE\_EMISSION * total\_volatile\_solids\ (kg)$$

[MN.ADG.1]

When:

- ACHIEVABLE_METHANE_EMISSION = achievable methane generation constant (m^3 CH_4 per kg VS loaded into digester); Constant Value: 0.24 m^3 CH_4/kg VS
- total_volatile_solids = daily mass (kg) of manure total volatile solids loaded into the digester, received from ManureStream(s)

Lastly, we need to convert daily CH_4 volume (generated_methane_volume) to CH_4 mass. First we calculate CH_4 density according to the user-provided digestion setpoint temperature, then we apply the density value to the volume of CH_4 generated.

$$methane\_density = \frac{METHANE\_MOLAR\_MASS}{(IDEAL\_GAS\_LAW\_R * (temperature\_set\_point + CELSIUS\_TO\_KELVIN)}$$

[MN.ADG.2]

When:

- METHANE_MOLAR_MASS (16.04 g/mol): molar mass of CH4
- IDEAL_GAS_LAW_R (0.0821 L atm/mol K): ideal gas law R value
- temperature_set_point ($°C$): user-provided digestion set point temperature
- CELSIUS_TO_KELVIN (273.15): value to convert $°C$ temperature values to K

$$generated\_methane\_mass\ (kg) = generated\_methane\_volume\ (m^3) * methane\_density$$

[MN.ADG.3]

When:

- generated_methane_volume (m^3): volume of CH_4 generated in the digester on a single day, calculated in **[MN.ADG.1]**.

Calculate destroyed volatile solids

_destroy_volatile_solids

Description: Microbes convert (destroy) VS during anaerobic digestion and produce biogas containing primarily CH_4 and CO_2. Therefore, the quantity of VS is assumed to be equal to the mass of CH_4 and CO_2 generated. In this section, we calculate the total mass of CO_2 and CH_4 generated, which is used later to update the degradable and non-degradable VS values of the ManureStream instance passed to the next processor. First, we calculate the density of CO_2 based on the user-provided digestion setpoint

temperature.

$$carbon\_dioxide\_density = \frac{CARBON\_DIOXIDE\_MOLAR\_MASS}{(IDEAL\_GAS\_LAW\_R * (temperature\_set\_point + CELSIUS\_TO\_KELVIN))}$$

[MN.ADG.4]

When:

- CARBON_DIOXIDE_MOLAR_MASS (44.01 g/mol): molar mass of CO_2
- IDEAL_GAS_LAW_R (0.0821 L atm/mol K): ideal gas law R value
- temperature_set_point ($°C$): user-provided digestion set point temperature
- CELSIUS_TO_KELVIN (273.15): value to convert $°C$ temperature values to K

Second, we determine the quantity of CO_2 volume and mass generated, assuming digester biogas contains a 60:40 volumetric ratio of CH_4 to CO_2.

$$generated\_carbon\_dioxide\_mass = (generated\_methane\_volume * CARBON\_DIOXIDE\_TO\_METHANE\_RATIO * CARBON\_DIOXIDE\_DENSITY)$$

[MN.ADG.5]

When:

- CARBON_DIOXIDE_MOLAR_MASS (44.01 g/mol): molar mass of CO_2; Constant Value: = 4/6 L/L (0.667)
- carbon_dioxide_density (kg per m^3) = ideal gas law value to convert CO_2 from mass to volume

Third, we calculate the total destruction of total VS.

$$total\_volatile\_solids\_destruction = generated\_methane\_mass + generated\_carbon\_dioxide\_mass$$

[MN.ADG.6]

We then utilize the value of total_volatile_solids_destruction (kg) to update VSd, manure VSnd, and bedding VSnd. As mentioned above, the total quantity of VS destroyed is partitioned between the three VS fractions according to the proportion of each fraction in manure entering the digester. E.g., if manure entering contained 60% VSd, 10% manure VSnd, and 30% bedding VSnd, 60% of destroyed VS will be subtracted from VSd, 10% from manure VSnd, and 30% from bedding VSnd. To do this, we

calculate the ratio of VSd to VS (degradable_volatile_solids_frac) and manure VSd to VS and apply these fractions to the total_volatile_solids_destruction value to determine the updated VS fraction values.

$$degradable\_VS\_frac = \frac{degradable\_VS\ (kg)}{total\_VS\ (kg)}$$

[MN.ADG.7]

$$manure\_non\_degradable\_VS\_frac = \frac{manure\_non\_degradable\_VS\ (kg)}{total\_VS\ (kg)}$$

[MN.ADG.8]

$$degradable\_VS = degradable\_VS\ (kg) - (total\_VS\_destruction\ (kg) * degradable\_VS\_frac)$$

[MN.ADG.9]

$$manure\_non\_degradable\_VS = manure\_non\_degradable\_VS\ (kg) - total\_VS_destruction * manure\_non\_degradable\_VS\_frac$$

[MN.ADG.10]

$$bedding\_non\_degradable\_VS = bedding\_non\_degradable\_VS\ (kg) - total\_VS_destruction * (1 - manure\_non\_degradable\_VS\_frac\ + degradable\_VS\_frac)$$

[MN.ADG.11]

Calculate methane leakage

_calculate_methane_leakage

Description: Calculates the volume of CH_4 generated that is lost to the atmosphere via leakage. The leakage fraction is currently a user input with a default value of 1%, which represents a conservative leakage rate. Leakage is largely dependent on the age and type of digester and should ideally be provided by the user for specificity.

$$methane\_leakage\_volume\ (m^3) = generated\_methane\_volume\ (m^3) * biogas\_leakage\_fraction$$

[MN.ADG.12]

When:

- generated_methane_volume (m^3): volume of CH_4 generated in the digester on a single day, calculated in **[MN.ADG.1]**
- biogas_leakage_fraction (%): fraction of biogas generated in the anaerobic digester that escapes to the atmosphere through unintended leakage; Default Value: 0.01 (1%)

Update digester effluent composition

_report_anaerobic_digester_outputs

The calculations below are some additional prerequisites to generating anaerobic digestion-specific outputs.

- The equation below is used to calculate the quantity of net, captured gas according to the quantity of biogas leakage.

$$captured\_methane\_volume\ (m^3) = generated\_methane\_volume\ (m^3) - methane\_leakage\_volume\ (m^3)$$

[MN.ADG.13]

When:

– generated_methane_volume (m^3): total volume of CH_4 generated in the digester on a specific day
– methane_leakage_volume (m^3): the volume of CH_4 generated that is lost to the atmosphere via leakage

- The equation below is used to calculate the updated volume of manure in the digester, accounting for the volume of destroyed VS.

$$updated\_volume(m^3) = incoming\_volume(m^3) - (\frac{total\_volatile\_solids\_destruction(kg)}{ManureConstants.SLURRY\_MANURE\_DENSITY})$$

[MN.ADG.14]

When:

– updated_volume (m^3) = incoming_volume (m^3) - (total_volatile_solids_destruction (kg) / ManureConstants.SLURRY_MANURE_DENSITY)

- Microbial processes during anaerobic digestion are known to increase the proportion/mass of total ammoniacal N (TAN), though the total mass of N is generally not different pre and post-digestion (Aguirre-Villegas, R. A. Larson, and Sharara, 2019). To reflect this, the proportion of TAN in manure in the digester is multiplied by a fixed factor. The factor (TAN_INCREASE_FACTOR, 1.60) was chosen based on a target of 50% loss of total N via NH_3-N emissions from a subsequent digestate lagoon, as reported by the USDA GHG estimation guidelines for uncovered digestate storage (Hanson, Itle, and Edquist, 2024).

$$updated\_ammoniacal\_nitrogen(kg) = min(ammoniacal\_nitrogen(kg) * TAN\_INCREASE\_FACTOR, nitrogen(kg))$$

[MN.ADG.15]

When:

– ammoniacal_nitrogen (kg): the mass of ammoniacal N in manure loaded into the digester on a single day
– TAN_INCREASE_FACTOR: factor by which total ammoniacal nitrogen content is increased by the anaerobic digestion process, set to 1.60
– Note that the “min” notation prevents manure TAN from exceeding manure total N.

C. Manure composition update

Below is a summary of updates to ManureStream variables. Note that the formulas below may be a summarization of multiple steps detailed above, and are intended to provide an overview of what mass losses/gains are reflected in the value of each variable. Equations in the table below are in the format of “Output/exiting value” = “Entering processor value” +/- XYZ. “Entering” refers to the manure entering the processor (i.e., being loaded into the digester) and output variables reflect the manure leaving the processor (i.e., digestate leaving the digester). The Calculation column describes how the variable is updated in the processor.

Table 17: Calculated anaerobic digestion outputs.

Variable	Units	Calculation
water	kg	Entering water (kg)
total_ammoniacal_nitrogen	kg	min(entering ammoniacal_nitrogen (kg) × TAN_INCREASE_FACTOR, nitrogen (kg))
nitrogen	kg	Entering nitrogen (kg)
phosphorus	kg	Entering phosphorus (kg)
potassium	kg	Entering potassium (kg)
ash	kg	Entering ash (kg)
degradable_volatile_solids		Entering degradable_volatile_solids (kg) − total_VS_destruction (kg) × degradable_VS_frac
manure_non_degradable_-volatile_solids		Entering non_degradable_volatile_solids (kg) − total_VS_destruction (kg) × manure_non_degradable_VS_frac
bedding_non_degradable_-volatile_solids		Entering non_degradable_volatile_solids (kg) − total_VS_destruction (kg) × (1 - degradable_VS_frac - manure_non_degradable_VS_frac)
total_solids	kg	Entering total_solids (kg) − total_VS_destruction (kg)
volume	m^3	Entering volume (m^3) − $\frac{\text{total\_VS\_destruction (kg)}}{\text{ManureConstants.SLURRY\_MANURE\_DENSITY}}$

VI. Slurry Storage

A. Introduction

Manure that is stored and managed at 7 to 12% dry matter is generally considered to be "slurry" manure. Several common options exist for storing manure at this %DM range. Slurry may be stored in underfloor pits, where manure is deposited directly into the pit through slatted floors, or is moved to an underfloor storage via scrapers or other manure handling systems. Manure may also be transported to outdoor storage tanks or pits/basins, which may be covered or uncovered.

Compared to anaerobic lagoons, slurry storages are typically of a smaller capacity, and do not result in controlled treatment (e.g., reduction of odor, N content reduction, organic matter decomposition) of manure. Slurry storages are typically emptied more frequently than liquid manure storages like anaerobic lagoons and contain more concentrated manure. In general, the biological processes in slurry storages and anaerobic lagoons are similar: microbes break down manure carbohydrates and proteins, resulting in CO_2, CH_4, and N_2O emissions, and mineralization of organic to inorganic N, and N losses occur through NH_3 volatilization at the manure surface. However, management of these two types of liquid manure storages differs as described above, which results in differences in emissions and nutrient losses.

Implementation in RuFaS

There are two options for slurry storage in RuFaS, slurry storage outdoor and slurry storage underfloor, which function almost identically. The key differences between the two methods are presented in Table 13. Because the two methods are highly similar, both are covered in this document.

Table 18: Key differences for two options of slurry storage.

	Slurry Storage Underfloor	**Slurry Storage Outdoor**
Temperature	Barn temperature method	Modeled air temperature method
Cover options	Only "no cover" permitted	Cover, cover and flare, crust, or no cover
Precipitation volume	Precipitation excluded from storage	Precipitation excluded or included depending on cover option selection

In the slurry storage submodules, accumulated manure in storage is modeled on a daily timestep. Nutrient/mass gains from daily addition of manure (feces/urine, bedding, wash water) to storage, and precipitation volume entering storage, are tracked. Gas emissions are calculated daily based on the quantity of nutrients in stored manure, manure temperature, storage type, use of a cover, and storage duration. Manure composition is then updated according to net nutrient losses/gains. Manure accumulates in storage until the end of the user-defined storage interval is reached. However, quantities of manure may additionally be removed from storage according to the user-defined manure application schedule. Note that at this time, water and nutrients from surface runoff, and water evaporation from storage manure, are not captured in slurry storage submodules.

Classes

Note that both slurry storage methods inherit some behavior from the base class, Storage. Because the two methods are highly similar, the content below is representative of both types of slurry storage, with specific differences between the two noted explicitly (see Table 18).

Slurry Storage	Description
SlurryStorageOutdoor(Storage)	slurry_storage_outdoor.py
SlurryStorageUnderfloor(Storage)	slurry_storage_underfloor.py

Required User Inputs

Table 19: Required inputs for the slurry storage section.

Variable	Units	Description
Name	none	Unique identifier of the specific slurry storage configuration used.
Capacity	m^3	The volumetric capacity of the slurry storage, in m^3. Note that this variable is a placeholder at this time, and does not influence model calculations.
Cover	-	Fefault value is "no cover" for slurry storage underfloor, as these storages do not generally have a synthetic cover, and typically receive too much surface disturbance to form a crust. Note that precipitation is always excluded from slurry storage underfloor given they are assumed to be completely covered or indoors. Default value for slurry storage outdoor is "no cover". Cover (represents either a synthetic, precipitation-excluding cover, or an enclosed tank) or crust (a naturally-forming crust over >50% of the stored manure surface) are also options.
Surface_area	m^2	The surface area of the slurry storage. If not provided by the user, surface area is calculated based on the number of mature cows in the herd; see Calculate Surface Area section below.
Storage_time_period	days	The number of days that manure is stored between emptying events. At the end of this interval, the manure storage is emptied completely.

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane production potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen

- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids
- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids
- total_solids
- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)
- volume
- methane_production_potential

Expected Outputs

- ManureStream variables representing manure loaded (received) into storage each day, and accumulated manure after accounting for nutrient and mass gains/losses
- storage_methane (kg): Daily emission of CH_4 from accumulated manure in slurry storage
- storage_ammonia (kg): Daily emission of NH_3 from accumulated manure in slurry storage
- storage_nitrous_oxide (kg): Daily emission of N_2O from accumulated manure in slurry storage.

B. Methodology

Calculate manure temperature
Slurry storage underfloor
_determine_barn_temperature

Temperature of stored manure is assumed to be equal throughout the entire mass of manure. In slurry storage underfloor, manure temperature is assumed to be equal to air temperature, but is bounded to 5 to $30°C$. See barn temperature determination method in Manure Handler section for more information.

Slurry storage outdoor
_determine_outdoor_storage_temperature

Daily temperature of manure in slurry storage outdoors is determined using a sine/cosine least squares fit to user-provided weather data, with a fixed amplitude damping and lag (phase shift) factor. For more information, see the *Calculate Stored Manure Temperature* in the Anaerobic Lagoon section.

Calculate storage surface area

Exposed surface area (m^2) of the manure in storage is important in determining NH_3-N emissions, as well as in determining precipitation volume added to storage if the storage is not covered or indoors. Wherever possible, this value should be provided by the user if modeling a real farm. If farm-specific information is unavailable or the farm being modeled is theoretical, the surface area should be estimated using tools like the USDA's Animal Waste Management Version 2.4.1. However, the RuFaS team recognizes that minimizing required inputs is desirable, though a fixed storage surface area is undesirable due to the variability in storage structure size and surface area. With this, an equation was developed that estimates storage surface area based on the following assumptions:

- All manure excreted by animals on the farm enters the specified storage. At this time, the Manure module is not capable of assessing the proportion of manure excreted that is stored in the defined storages, therefore, all manure is assumed to be stored in the current storage, for the purposes of surface area estimation.
- The storage is 15 ft deep, with vertical walls.
- The storage receives 2500 mm of precipitation per year.
- Herd composition, and thus manure excretion, is fixed, and the number of animals in each life stage class is proportional to the number of mature cows.

A constant value was derived to calculate estimated manure excretion based on the number of mature cows housed on the farm (a user input). The average number of animals in each class was determined according to default RuFaS animal lifecycle inputs, and the total mass and volume of manure excreted by the herd was calculated. This resulted in an estimated daily herd-wide manure excretion of 168.6 kg or 0.118 m^3 of manure per mature cow housed on the farm. The resulting equation is used to calculate storage surface area (m^2).

$$surface\_area(m^2) = \frac{(cow\_num * MANURE\_CONVERSION\_CONSTANT \times storage\_time \times FREEBOARD\_CONSTANT)}{(DEPTH\_CONSTANT - PRECIPITATION\_CONSTANT)}$$

[MN.STO.1]

When:

- cow_num: user-inputted number of mature cows housed on the farm
- MANURE_CONVERSION_CONSTANT: Factor to estimate m^3 of herd-wide manure produced per day per mature cow housed on the farm, set to 0.1175 m^3.
- storage_time (days): user-inputted number of days that manure is stored in this storage for before being emptied.
- FREEBOARD_CONSTANT: the volume allowance above the maximum volume of a slurry or liquid manure storage, set to 1.20 (20%).
- DEPTH_CONSTANT: value for slurry or liquid manure storage depth, set to 4.572 m (15 feet).
- PRECIPITATION_CONSTANT: The annual precipitation constant value, used only in determination of storage surface area if surface area is not provided by the user, set to 0.25 m.

Calculate precipitation volume

Covers have implications for inclusion or exclusion of precipitation volume, as well as for N_2O emissions. Four cover options exist for slurry storages:

- “Cover”: An impermeable cover that does not permit precipitation to enter the storage.
- “Cover and flare”: An impermeable cover with flaring of methane produced in storage. See Cover and Flare section.
- “Crust”: A naturally forming crust exists on the surface of the slurry storage.
- “No cover”: Storage is not covered or indoors.

The cover type for slurry storage underfloor in the default manure management file is "uncovered", as these storages are typically not enclosed. However, precipitation is always excluded from underfloor slurry storages, regardless of the cover type – see Precipitation below.
Precipitation volume for uncovered outdoor slurry storages is calculated as follows:

$$Daily\ precipitation\ volume(m^3) = storage\ surface\ area(m^2) \times precipitation(m)$$

[MN.STO.2]

When:

- Storage surface area: the user-defined or model-estimated storage surface area (m^2).
- Precipitation: the daily amount of precipitation (m).

Calculate methane emissions

_calculate_methane_emissions

We use an adaptation of a method originally conceived by Sommer, Petersen, and Moller, 2004 to calculate daily emissions of CH_4 from degradable and non-degradable VS in slurry. These equations focus on the degradation of degradable and non-degradable volatile solids (VS) present in the manure. Factors like degradable and non-degradable VS (VSd and VSnd) content in storage, temperature, and location (indoor/outdoor) affect estimated CH_4 emissions. We apply the original method from Sommer, Petersen, and Moller, 2004 with updated dairy manure Arrhenius and activation energy values from Elsgaard, Olsen, and Petersen, 2016 and Petersen et al., 2024.

Methane emissions are calculated in the same way for slurry storage outdoor and underfloor, with the exception that temperature of manure in outdoor vs. underfloor slurry storage is determined differently, as described below. The same equation is utilized to calculate CH_4 emissions from VSd and VSnd (from both manure and bedding sources), though the rate-correcting factor differs between the two.

First, we must calculate the value of the Arrhenius exponent (_calculate_arrhenius_exponent). This value directly represents the responsiveness of biological reaction speed to temperature, and in the context of this empirical equation, may also be related to the methane potential of manure in storage and activity of the microbial population:

$$Arrh\_exp\ g\ (CH_4\ kg^{-1}\ VS\ h^{-1}) = e^{Ln(A) - \frac{ACTIVATION\_ENERGY}{(Gasconstant * manuretemperature)}}$$

[MN.MET.2]

When:

- Ln(A): The natural log of the Arrhenius parameter (NATURAL_LOG_ARRHENIUS_CONSTANT constant), set at 30.7 based on Petersen et al., 2024. This is an empirically-derived value determined based on observed manure CH_4 emission values.
- ACTIVATION_ENERGY: the apparent activation energy of methanogenesis in cattle slurry (J/mol), set at 81,000 J/mol, based on Elsgaard, Olsen, and Petersen, 2016.
- Gas constant: ideal gas constant, set at 8.314 J K/mol.
- Manure temperature (K): temperature of manure in storage.

Now we can calculate actual daily CH_4 emission, based on the total quantity of VSd and VSnd in stored manure. The basic equation, used to calculate CH_4 emissions for each VS fraction, is as follows:

$$CH_4\ emission\ from\ VS_{d\ or\ nd}\ (kg\ d^{-1}) = 24 * Arrh\_exp * VS_{d\ or\ nd} * rate\_factor$$

[MN.MET.3]

When:

- 24: conversion factor from hours to day.
- Arrh_exp: Arrhenius parameter for CH_4 emission rate (g CH_4 kg^{-1} VS h^{-1}), calculated in **[MN.MET.2]**.
- $VS_{d\ or\ nd}$: The mass (kg) of VSd or VSnd in manure in slurry storage.
- rate_factor: The unitless rate-correcting factor, set to 1 for VS_d and 0.01 for VS_{nd}.

The total daily CH_4 emission is equal to the sum of emissions from the VS_d and VS_{nd} fractions.

Calculate cover and flare methane emissions

_calculate_cover_and_flare_methane

The cover and flare option is applicable to slurry storage outdoor only (i.e., not usable with slurry storage underfloor). If the cover and flare option is selected, daily CH_4 emission from slurry storage is multiplied by a methane destruction efficiency value. The set value for methane destruction efficiency is 81%, based on a white paper commissioned by Dairy Management, Inc. on cover and flare efficiency (**wallaceDMI**). The updated daily CH_4 emission (kg) from a cover and flare slurry storage is as follows:

$$Daily\ storage\ CH_4\ (kg) : storage\ CH_4 \times (1 - METHANE\_DESTRUCTION\_EFFICIENCY)$$

[MN.MET.4]

When:

- Storage CH_4 (kg): total daily kg of CH_4 emitted from stored manure, calculated in **[MN.MET.3]**.
- METHANE_DESTRUCTION_EFFICIENCY: coefficient for destruction of methane by the flare, set to 0.81.

Calculate volatile solids losses

_apply_methane_emissions

Daily emissions of CH_4 and CO_2 from slurry storage occur through microbial degradation of VS in slurry manure, among other processes (Petersen et al., 2024). Therefore, gaseous emissions from slurry storage result in a decrease in the quantity of VS in stored slurry. VSd and VSnd remaining in manure are updated separately according to their respective loss via CH_4 **[MN.STO.4]**. Here, we assume a fixed 1:3 molar ratio of CH_4-C to CO_2-C emissions from stored slurry from Petersen et al., 2024. This enables calculation of the total amount of C and thus VS_d and VS_{nd} lost through CH_4 and CO_2 emissions based on the quantity of CH_4 emitted from each VS fraction.

Given that C is assumed to be lost via CH_4 and CO_2 emissions in a ratio of 1:3, we assume for each C lost as CH_4, 3 C are lost as CO_2. CH_4 is 75% C by mass, thus for each kg of CH_4 emitted, 0.7498 C are lost via CH_4 and (3 x 0.7498) are lost from CO_2, for a total of 2.992 kg C per kg of CH_4 emitted. We assume manure VS are 45% C (Petersen et al., 2024); therefore, 2.9992 kg C / 45% C = 6.665 kg VS are lost per kg of CH_4 emitted.

$$VS_{d\ or\ nd}\ loss\ (kg) = \ CH_4\ emission\ from\ VS_{d\ or\ nd}\ (kg) * \ VS\_TO\_METHANE\_LOSS\_RATIO$$

[MN.STO.3]

When:

- CH_4 emission from VS_d or $_{nd}$ (kg): total daily kg of CH_4 emitted from VS_d or VS_{nd}, calculated in **[MN.MET.3]**
- VS_TO_METHANE_LOSS_RATIO: default ratio of VS degraded per kg of CH_4 emitted from slurry storage, set to 6.665

Calculate ammonia emissions
_calculate_ammonia_emissions

Emission of NH_3-N from stored slurry is determined using equations from Rotz and Oenema, 2006, which are also utilized in the IFSM (Rotz, Corson, et al., 2023). Ammonia emissions are influenced by the quantity of total ammoniacal N (TAN) accumulated in manure storage, manure temperature, and manure storage surface area. First, we must derive the various parameters utilized in the calculation.

- First, we need to derive the value of the equilibrium coefficient Q for the NH_3 gas in the air for a given concentration of TAN in stored manure using Henry's law. Note that the concentration of NH_3 in the free atmosphere is assumed to be zero. Since Q is a function of the Henry's law coefficient K_h and a dissociation of ammonium coefficient K_a, we will calculate those first.

  Henry's law coefficient (K_h)

$$K_h = 10^{\frac{1478}{manure\ temperature}} - 1.69$$

[MN.AMM.1]

  When:

  – Manure temperature (K): temperature of manure storage.

  Dissociation coefficient of ammonium (K_a)

$$K_a = 1 + 10^{(0.09018 + \frac{2729.9}{manure\ temperature} - pH)}$$

[MN.AMM.2]

  When:

  – Manure temperature (K): temperature of stored manure.
  – DEFAULT_STORED_MANURE_PH: the pH of the manure in storage, set to 7.5 by default

Equilibrium coefficient (Q)

$$Q = K_h * K_a$$

[MN.AMM.3]

When:

- K_h: Henry's law coefficient, calculated in **[MN.AMM.1]**.
- K_a: Dissociation coefficient of ammonium, calculated in **[MN.AMM.2]**.

- Next, the rate of NH_3-N loss in kg N/m^2 from stored manure is calculated:

$$NH_3N\ emission\ rate\ (kg\ N/m^2) = \frac{(TAN*c*y)}{(STORAGE\_RESISTANCE*M*Q)}$$

[MN.AMM.5]

When:

- TAN (kg): Mass of ammoniacal N in stored manure
- c: time conversion constant (86400 s per d)
- y: manure density, set to 990 kg/m^3
- STORAGE_RESISTANCE: A constant value representing the sum of resistance of NH_3 transfer from solution to manure surface, and from manure surface to atmosphere, set at 23.1 s/m.
- M (kg): Total mass of stored manure
- Q: Equilibrium coefficient calculated in **[MN.AMM.3]**

- Lastly, we calculate total NH_3-N emissions (kg), based on the emission rate we just calculated and the manure storage surface area.

$$NH_3N\ emissions\ (kg) = NH_3N\_rate * surface\_area$$

[MN.AMM.7]

When:

- NH_3N_rate (kg N/m^2): Rate of NH_3-N loss (kg/m^2) from manure, calculated in **[MN.AMM.5]**.
- surface_area (m^2): Total manure storage surface area.

Calculate nitrous oxide emissions

_calculate_nitrous_oxide_emissions

N_2O emissions (kg N_2O-N) are based on the daily quantity of manure N loaded into storage, whether the manure storage is covered or uncovered. This method is based on Intergovernmental Panel on Climate Change, 2019; however, it should be noted that the original IPCC, 2006 method is based on daily manure N excretion by animals, whereas the current method is based on manure N loading into storage, which may reflect upstream N losses from NH_3 emissions in housing, solid liquid separation, etc. The calculation is as follows:

$$N_2O - N\ emissions\ (kg) = Received\_N * N_2O\ factor$$

[MN.NIT.1]

When:

- Received_N (kg): Quantity of manure total N loaded into storage on the current day
- N_2O factor: kg of N_2O-N emitted per kg of manure N added per day to storage, based on the following logic:
  - Cover type = crust OR cover; 0.005
  - Cover type = no cover; 0 (no N_2O emissions)

C. Received, stored, and emptied outputs

Manure storages in RuFaS report two types of outputs to OutputManager each day: received manure and stored manure.

Received manure outputs represent the quantity of manure mass and nutrients added to the manure storage on a single day. No nutrient losses from gas or other emissions/losses are reflected in these output values.

Stored manure outputs represent the accumulated quantity of manure and nutrients present in storage on a single day. These values are the net quantity of mass/nutrients remaining each day after adding received manure values and subtracting any losses to gas emissions or other losses. In slurry storage processors, daily losses include CH_4, NH_3, and N_2O emissions. The order of operations in updating accumulated manure values is:

- Add received manure values to stored manure values
- Calculate gas emissions and total nutrient losses based on stored manure values
- Update stored manure values based on the day's nutrient losses. See the Manure composition update section for specific details on how nutrient gains and losses are accounted for on a daily timestep.

For slurry storage and all other storage processor types, the stored manure values (not received manure) are passed to the next processor in the chain (e.g. another storage, field application, export, etc.) when the storage time interval is complete.

Emptied manure

Manure may be removed from storage via requests made by the Crop and Soil module. The user specifies the days and years for manure removal (i.e. application), as well as the application type (liquid or solid) and quantity of N or P required for each application date within year. Note that these actions are the responsibility of the Crop and Soil module; more information on manure application inputs and methodology can be found in the Crop and Soil module documentation. When manure is removed from storage by the Crop and Soil module, emptied manure outputs report the quantity of manure and nutrients removed on that day, and Manure Stream attributes representing stored manure are updated accordingly to reflect post-removal amounts remaining in storage.

D. Manure composition updates

Received manure:

In slurry storage processors, the following nutrient sources are represented in received manure values:

- ManureStream values, as received from the previous processor(s) in the manure management chain
- Precipitation water (kg), calculated in MN.STO.2 (if applicable), is added to the water value in ManureStream

Stored manure:

Below is a summary of updates to ManureStream variables representing the stored manure. Note that the formulas below may be a summarization of multiple steps detailed above, and are intended to provide an overview of what mass losses/gains are reflected in the value of each variable.

Equations in the table below (Calculation column) are in the format of: updated stored manure value = yesterday's stored manure value + today's manure value +/- XYZ. The updated stored manure values reflect the total quantity of manure/nutrients in storage on a single day after accounting for all gains/losses that occurred on that day. Received manure simply refers to the manure being loaded into the manure storage each day.

Table 20: Manure storage variable calculations

Variable	Units	Calculation
water	kg	stored manure water (kg) + received manure water (kg)
total_ammoniacal_nitrogen	kg	max(0, stored manure ammoniacal nitrogen (kg) + received ammoniacal nitrogen (kg) $-NH_3N$ emissions (kg))
nitrogen	kg	stored manure nitrogen (kg) + received manure nitrogen (kg) − NH_3N emissions (kg) − N_2O-N emissions (kg)
phosphorus	kg	stored manure phosphorus (kg) + received manure phosphorus (kg)
potassium	kg	stored manure potassium (kg) + received manure potassium (kg)
ash	kg	stored manure ash (kg) + received manure ash (kg)
degradable_volatile_solids		stored degradable VS (kg)+received degradable VS (kg)− VSd loss (kg)
manure_non_degradable_-volatile_solids		stored manure non-degradable VS (kg) + received manure non-degradable VS (kg) − manure VSnd loss (kg)
bedding_non_degradable_-volatile_solids		stored bedding non-degradable VS (kg) + received bedding non-degradable VS (kg) − bedding VSnd loss (kg)
total_solids	kg	stored total solids (kg) + received total solids (kg) − VSd loss (kg) − VSnd loss (kg)
volume	m^3	stored volume (m^3) + Received volume (m^3) − $\frac{\text{VSd loss (kg)+VSnd loss (kg)}}{\text{SLURRY\_MANURE\_DENSITY}}$

VII. Anaerobic Lagoon

A. Introduction

Manure that is stored and managed at less than $\tilde{5}$% dry matter is generally considered to be liquid manure. Liquid manure is generated by either dilution of raw or slurry manure, generally through the addition of wash or flush water, or removal of a portion of manure solids through solid liquid separation methods (mechanical separator, settling basin, etc.) or anaerobic digestion. Liquid manure is generally stored in a type of large, outdoor storage structure called an anaerobic lagoon, or simply a lagoon. Anaerobic lagoons are not simply structures in which to store manure. Lagoons facilitate biological breakdown of organic materials, which reduces volatile solids content and odor, though also increases N mineralization and loss as ammonia, particularly if the lagoon is uncovered. Accordingly, anaerobic lagoons have specific design and management requirements to facilitate biological treatment activity (Natural Resources Conservation Service (NRCS), 2017). Some characteristics that separate an anaerobic lagoon from slurry or liquid manure storage are:

- Greater storage capacity
- Less frequent and less complete emptying, resulting in longer solids/sludge retention time
- Storage of liquid rather than slurry manure
- Controlled volatile solids loading rate
- Lagoons are generally a lined or unlined in-ground basin, whereas slurry storage may be either in-ground or above-ground tanks or other structures

Implentation in RuFaS

In RuFaS, the underlying biological and gas emission methods are identical for slurry storages vs. anaerobic lagoons, as the biological process of organic matter breakdown is very similar between the two in reality. However, the differences in size, dilution, management, and other factors differ between the two in reality, leading to generally greater GHG emissions from lagoons compared to slurry storages. In the anaerobic lagoon submodule, accumulated manure in storage (i.e., held in the lagoon) is modeled on a daily timestep. Nutrient/mass gains from daily addition of manure (feces/urine, bedding, wash water) to storage, and precipitation volume entering storage, are tracked. Gas emissions are calculated daily based on the quantity of nutrients in stored manure, manure temperature, storage type, use of a cover, and storage duration. Manure composition is then updated according to net nutrient losses/gains. Manure accumulates in storage until the end of the user-defined storage interval is reached. However, quantities of manure may additionally be removed from storage according to the user-defined manure application schedule

Classes

Table 21: List of Python classes for anaerobic lagoon.

Anaerobic Lagoon	Description
AnaerobicLagoon(Storage)	anaerobic_lagoon.py

Required User Inputs

Table 22: Required inputs for the anaerobic lagoon section (refreshed_manure_management.json)

Variable	Units	Description
Name	none	Unique identifier of the specific anaerobic lagoon configuration used.
Capacity	m^3	The volumetric capacity of the anaerobic lagoon, in m^3. Note that this variable is a placeholder at this time, and does not influence model calculations.
Cover	-	The type of cover used with the anaerobic lagoon.
Surface_area	m^2	The surface area of the anaerobic lagoon at the minimum operating level.
Storage_time_-period	days	The number of days that manure is stored between emptying events. At the end of this interval, the manure storage is emptied completely.

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane production potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen
- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids
- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids
- total_solids
- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)

- volume
- methane_production_potential

Expected Outputs

- ManureStream variables representing manure loaded (received) into storage each day, and accumulated manure after accounting for nutrient and mass gains/losses
- storage_methane (kg): Daily emission of CH_4 from accumulated manure in an anaerobic lagoon.
- storage_ammonia (kg): Daily emission of NH_3 from accumulated manure an anaerobic lagoon.
- storage_nitrous_oxide (kg): Daily emission of N2O from accumulated manure an anaerobic lagoon.

B. Methodology

Calculate manure temperature

_determine_outdoor_storage_temperature

Manure temperature is modeled using a cosine function whose parameters are derived from a least-squares fit of simulation-wide weather data. The air temperature amplitude is reduced using a damping factor to reflect the smaller annual variation in manure temperature relative to air. The phase shift (i.e., timing of peak temperature) is determined based on the least squares function and is adjusted by a fixed lag constant representing the delayed thermal response of manure temperature relative to air temperature.

First we determine the amplitude of the manure temperature function by applying the damping factor.

$$manure\_amplitude = amplitude \times MANURE\_DAMPING\_FACTOR$$

[MN.STO.13]

When:

- amplitude: modeled amplitude of the seasonal air temperature function, calculated from user-supplied, simulation-wide weather data
- MANURE_DAMPING_FACTOR: a fixed damping factor applied to the air temperature amplitude, set to 0.65.

Second, we use this amplitude in the following function to determine modeled manure temperature(°C) each simulation day. Note the function includes a 'max' term to implement a lower temperature bound for manure temperature.

$$manure\_temp = max((mean\_temp \times manure\_amplitude \times \cos \frac{2\pi}{365} \times (jday - phase\_shift - MANURE\_TEMPERATURE\_LAG)), min\_temp)$$

[MN.STO.14]

When:

- mean_temp (°C): simulation-wide mean air temperature
- manure_amplitude: amplitude of the manure temperature function, determined in **MN.STO.13**
- jday: Julian day of the simulation
- phase_shift: Julian date of peak air temperature in the simulation

- MANURE_TEMPERATURE_LAG (days): fixed lag constant representing the delayed thermal response of manure temperature relative to air temperature, set to 30.
- min_temp (°C): A fixed minimum manure temperature constant, dependent on the type of storage.
  - Anaerobic lagoon: 1 °C
  - Slurry storage outdoor: -20 °C

Calculate storage surface area

Exposed surface area (m^2) of the manure in storage is important in determining NH_3-N emissions, as well as in determining precipitation volume added to storage if the storage is not covered or indoors. Wherever possible, this value should be provided by the user if modeling a real farm. If farm-specific information is unavailable or the farm being modeled is theoretical, the surface area should be estimated using tools like Animal Waste Management. However, the RuFaS team recognizes that minimizing required inputs is desirable, though a fixed storage surface area is undesirable due to the variability in storage structure size and surface area. With this, an equation was developed that estimates storage surface area based on the following assumptions:

- All manure excreted by animals on the farm enters the specified storage. At this time, the Manure module is not capable of assessing the proportion of manure excreted that is stored in the defined storages, therefore, all manure is assumed to be stored in the current storage, for the purposes of surface area estimation.
- The storage is 15 ft deep, with vertical walls.
- The storage receives 2500 mm of precipitation per year.
- Herd composition, and thus manure excretion, is fixed, and the number of animals in each life stage class is proportional to the number of mature cows.

A constant value was derived to calculate estimated manure excretion based on the number of mature cows housed on the farm (a user input). The average number of animals in each class was determined according to default RuFaS animal lifecycle inputs, and the total mass and volume of manure excreted by the herd was calculated. This resulted in an estimated daily herd-wide manure excretion of 168.6 kg or 0.118 m^3 of manure per mature cow housed on the farm. The resulting equation is used to calculate storage surface area (m^2).

$$surface\_area(m^2) = \frac{(cow\_num * MANURE\_CONERSION\_CONSTANT * storage\_time * FREEBOARD\_CONSTANT)}{(DEPTH\_CONSTANT - PRECIPITATION\_CONSTANT)}$$

[MN.STO.1]

When:

- cow_num: user-inputted number of mature cows housed on the farm
- MANURE_CONVERSION_CONSTANT: Factor to estimate m^3 of herd-wide manure produced per day per mature cow housed on the farm, set to 0.1175 m^3.
- storage_time (days): user-inputted number of days that manure is stored in this storage for before being emptied
- FREEBOARD_CONSTANT: the volume allowance above the maximum volume of a slurry or liquid manure storage, set to 1.20 (20%).

- DEPTH_CONSTANT: value for slurry or liquid manure storage depth, set to 4.572 m (15 feet).
- PRECIPITATION_CONSTANT: The annual precipitation constant value, used only in determinatio

Calculate precipitation volume

The use of covers has implications for inclusion or exclusion of precipitation volume, as well as for N^2O emissions. Four cover options exist for anaerobic lagoons.

- Cover
- Cover and flare
- Crust
- No cover

Detailed descriptions are outlined in the Slurry Storage section of this module. Precipitation volume for anaerobic lagoons that are uncovered or have a crust is calculated as follows:

$$Daily\ precipitation\ volume\ (m^3) = storage\ surface\ area\ (m^2) *\ precipitation\ (m)$$

[MN.STO.2]

When:

- Storage surface area: the user-defined or model-estimated storage surface area (m^2).
- Precipitation: the daily amount of precipitation (m).

Calculate methane emissions

_calculate_methane_emissions

We use an adaptation of a method originally conceived by Sommer, Petersen, and Moller, 2004 to calculate daily emissions of CH_4 from degradable and non-degradable VS in anaerobic lagoons. These equations focus on the degradation of degradable and non-degradable volatile solids (VS) present in the manure. Factors like degradable and non-degradable VS (VSd and VSnd) content in storage, temperature, and location (indoor/outdoor) affect estimated CH_4 emissions. We apply the original method from Sommer et al., 2022 with updated dairy manure Arrhenius and activation energy values from Elsgaard, Olsen, and Petersen, 2016 and Petersen et al., 2024. The same equation is utilized to calculate CH_4 emissions from VSd and VSnd, though the rate-correcting factor differs between the two. First, we must calculate the value of the Arrhenius exponent (_calculate_arrhenius_exponent). This

value directly represents the responsiveness of biological reaction speed to temperature, and in the context of this empirical equation, may also be related to the methane potential of manure in storage and activity of the microbial population:

$$Arrh\_exp\ (g\ CH_4\ kg^{-1}\ VS\ h^{-1}) = e^{Ln(A) - \frac{ACTIVATION\_ENERGY}{(Gasconstant*manuretemperature)}}$$

[MN.MET.2]

When:

- Ln(A): The natural log of the Arrhenius parameter (NATURAL_LOG_ARRHENIUS_CONSTANT), set at 30.6 based on Petersen et al., 2024. This is an empirically-derived value determined based on observed manure CH_4 emission values.

- ACTIVATION_ENERGY: the apparent activation energy of methanogenesis in cattle slurry (J/-mol), set at 81,000 J/mol, based on Elsgaard, Olsen, and Petersen, 2016.
- Gas constant: ideal gas constant, set at 8.314 J K/mol.
- Manure temperature (K): temperature of manure in storage.

Now we can calculate actual daily CH_4 emission, based on the total quantity of VSd and VSnd in stored manure. The basic equation, used to calculate CH_4 emissions for each VS fraction, is as follows:

$$CH_4\ emission\ from\ VS_{d\ or\ nd}\ (kg\ d^{-1}) = 24 * Arrh\_exp * VS_{d\ or\ nd} * rate\_factor$$

[MN.MET.3]

When:

- 24: conversion factor from hours to day.
- Arrh_exp: Arrhenius parameter for CH_4 emission rate (g CH_4 kg^{-1} VS h^{-1}), calculated in **[MN.MET.2]**.
- S_d or $_{nd}$: The mass (kg) of VS_d or VS_{nd} (from both manure and bedding) in manure in storage.
- rate_factor: The unitless rate-correcting factor, set to 1 for VS_d and 0.01 for VS_{nd}.

The total daily CH_4 emission is equal to the sum of emissions from the VS_d and VS_{nd} fractions.

Calculate cover and flare methane emissions
_calculate_cover_and_flare_methane

If the cover and flare option is selected, daily CH_4 emission from an anaerobic lagoon is multiplied by a methane destruction efficiency value. The set value for methane destruction efficiency is 81%, based on a white paper commissioned by Dairy Management, Inc. on cover and flare efficiency authored by Jim Wallace and co-authors. The updated daily CH_4 emission (kg) from a cover and flare lagoon is as follows:

$$Daily\ storage\ CH_4\ (kg) : storage\ CH_4 \times (1 - METHANE\_DESTRUCTION\_EFFICIENCY)$$

[MN.MET.4]

When:

- Storage CH_4 (kg): total daily kg of CH_4 emitted from stored manure, calculated in **[MN.MET.4]**.
- METHANE_DESTRUCTION_EFFICIENCY: destruction efficiency: coefficient for destruction of methane by the flare, set to 0.81.

Calculate volatile solids losses
_apply_methane_emissions

Daily emissions of CH_4 and CO_2 from anaerobic lagoons occur through microbial degradation of VS in manure, among other processes (Petersen et al., 2024). Therefore, gaseous emissions from lagoons result in a decrease in the quantity of VS in stored manure. VSd and VSnd remaining in manure are updated separately according to their respective loss via CH_4 **[MN.STO.4]**. Here, we assume a fixed

1:3 molar ratio of CH_4-C to CO_2-C emissions from stored manure from Petersen et al., 2024. This enables calculation of the total amount of C and thus VS_d and VS_{nd} lost through CH_4 and CO_2 emissions based on the quantity of CH_4 emitted from each VS fraction.

Given that C is assumed to be lost via CH_4 and CO_2 emissions in a ratio of 1:3, we assume for each C lost as CH_4, 3 C are lost as CO_2. CH_4 is 75% C by mass, thus for each kg of CH_4 emitted, 0.7498 C are lost via CH_4 and (3 x 0.7498) are lost from CO_2, for a total of 2.992 kg C per kg of CH_4 emitted. We assume manure VS are 45% C (Petersen et al., 2024); therefore, 2.9992 kg C / 45% C = 6.665 kg VS are lost per kg of CH_4 emitted.

$$VS_{d\ or\ nd}\ loss\ (kg) = \ CH_4\ emission\ from\ VS_{d\ or\ nd}\ (kg) * \ VS\_TO\_METHANE\_LOSS\_RATIO$$

[MN.STO.3]

When:

- CH_4 emission from VSd or nd (kg): total daily kg of CH_4 emitted from VS_d or VS_{nd}, calculated in MN.MET.3
- VS_TO_METHANE_LOSS_RATIO: default ratio of VS degraded per kg of CH_4 emitted from slurry storage, set to 6.665

Calculate manure retention at emptying
_emptying_fraction

Anaerobic lagoons, through their settling action, accumulate and retain a bottom layer of solids often referred to as "sludge". Additionally, depending on the frequency and extent of lagoon agitation, retention time of volatile solids in lagoons is typically explicitly managed to promote biological degradation of solids. These factors contribute to the generally greater CH_4 emissions per unit of volatile solids loaded into anaerobic lagoons compared to in-ground basin or tank manure storages. To directly capture the greater retention of manure at emptying events, and to indirectly capture the greater biological activity in anaerobic lagoons, a default manure retention factor is implemented in RuFaS. This factor dictates the portion of manure which, when the storage time interval is reached, is retained in the lagoon. This factor is applied evenly to all manure constituents (i.e., ManureStream variables).

$$retained\_manure_i = \\ accumulated\_manure_i \times ANAEROBIC\_LAGOON\_MANURE\_RETENTION$$

[MN.STO.15]

When:

- i: manure consistituent i
- accumulated_manure_$_i$: quantity of manure consistituent i present in the accumulated anaerobic lagoon manure when the
- ANAEROBIC_LAGOON_MANURE_RETENTION: constant fraction of the accumulated stored manure that is retained in the anaerobic lagoon when the storage time interval is reached, set to 0.10

Calculate ammonia emissions
_calculate_ammonia_emissions

Emission of NH_3-N from anaerobic lagoons is determined using equations from Rotz and Oenema, 2006, which are also utilized in the IFSM (Rotz, Corson, et al., 2023). Ammonia emissions are influenced by the quantity of total ammoniacal N (TAN) accumulated in manure storage, manure temperature, and manure storage surface area. First, we must derive the various parameters utilized in the calculation.

- First, we need to derive the value of the equilibrium coefficient Q for the NH_3 gas in the air for a given concentration of TAN in stored manure using Henry's law. Note that the concentration of NH_3 in the free atmosphere is assumed to be zero. Since Q is a function of the Henry's law coefficient K_h and a dissociation of ammonium coefficient K_a, we will calculate those first.

  Henry's law coefficient (K_h)

  $$K_h = 10^{\frac{1478}{manure\ temperature}} - 1.69$$

  [MN.AMM.1]

  When:

  – Manure temperature (K): temperature of manure storage.

  Dissociation coefficient of ammonium (K_a)

  $$K_a = 1 + 10^{(0.09018 + \frac{2729.9}{manure\ temperature} - pH)}$$

  [MN.AMM.2]

  When:

  – Manure temperature (K): temperature of manure storage.
  – DEFAULT_STORED_MANURE_PH: the pH of the manure in storage, set to 7.5 by default

  Equilibrium coefficient (Q)

  $$Q = K_h * K_a$$

  [MN.AMM.3]

  When:

  – K_h: Henry's law coefficient, calculated in **[MN.AMM.1]**.
  – K_a: Dissociation coefficient of ammonium, calculated in **[MN.AMM.2]**.

- Next, the rate of NH_3-N loss in kg N/m2 from stored manure is calculated:

  $$NH_3N\ emission\ rate\ (kg\ N/m^2) = \frac{(TAN*c*y)}{(STORAGE\_RESISTANCE*M*Q)}$$

[MN.AMM.5]

When:

- TAN (kg): Mass of ammoniacal N in stored manure
- c: time conversion constant (86400 s per d)
- y: manure density, set to 990 kg/m^3
- STORAGE_RESISTANCE: A constant value representing the sum of resistance of NH_3 transfer from solution to manure surface, and from manure surface to atmosphere, set at 23.1 s/m.
- M (kg): Total mass of stored manure
- Q: Equilirbium coefficient calculated in **[MN.AMM.3]**

- Lastly, we calculate total NH_3-N emissions (kg), based on the emission rate we just calculated and the manure storage surface area.

$$NH_3N\ emissions\ (kg) = NH_3N\_rate * surface\_area$$

[MN.AMM.7]

When:

- NH_3N_rate (kg N/m^2): Rate of NH_3-N loss (kg/m^2) from manure, calculated in **[MN.AMM.5]**
- surface_area (m^2): total manure storage surface area

Calculate nitrous oxide emissions
_calculate_nitrous_oxide_emissions

N2O emissions (kg N2O-N) are based on the daily quantity of manure N loaded into the lagoon, and whether the lagoon is covered or uncovered. This method is based on Intergovernmental Panel on Climate Change, 2019; however, it should be noted that the original IPCC method is based on daily manure N excretion by animals, whereas the current method is based on manure N loading into storage, which may reflect upstream N losses from $_3$ emissions in housing, solid liquid separation, etc. The calculation is as follows:

$$N_2ON\ emissions\ (kg) = Received\_N * N_2O\ factor$$

[MN.NIT.1]

When:

- Received_N (kg): Quantity of manure total N loaded into storage on the current day
- N_2O factor: kg of N_2O-N emitted per kg of manure N added per day to storage, based on the following logic:
  - Cover type = crust OR cover; 0.005
  - Cover type = no cover; 0 (no N_2O emissions)

C. Received, stored, and emptied outputs

Manure storages in RuFaS report two types of outputs to OutputManager each day: received manure and stored manure.

Received manure outputs represent the quantity of manure mass and nutrients added to the manure storage on a single day. No nutrient losses from gas or other emissions/losses are reflected in these output values.

Stored manure outputs represent the accumulated quantity of manure and nutrients present in storage on a single day. These values are the net quantity of mass/nutrients remaining each day after adding received manure values and subtracting any losses to gas emissions or other losses. In anaerobic lagoon processors, daily losses include CH_4, NH_3, and N_2O emissions. The order of operations in updating accumulated manure values is:

- Add received manure values to stored manure values
- Calculate gas emissions and total nutrient losses based on stored manure values
- Update stored manure values based on the day's nutrient losses. See the Manure composition update section for specific details on how nutrient gains and losses are accounted for on a daily timestep.

For anaerobic lagoons and all other storage processor types, the stored manure values (not received manure) are passed to the next processor in the chain (e.g. another storage, field application, export, etc.) when the storage time interval is complete.

Emptied manure

Manure may be removed from storage via requests made by the Crop and Soil module. The user specifies the days and years for manure removal (i.e. application), as well as the application type (liquid or solid) and quantity of N or P required for each application date within year. Note that these actions are the responsibility of the Crop and Soil module; more information on manure application inputs and methodology can be found in the Crop and Soil module documentation. When manure is removed from storage by the Crop and Soil module, emptied manure outputs report the quantity of manure and nutrients removed on that day, and Manure Stream attributes representing stored manure are updated accordingly to reflect post-removal amounts remaining in storage.

D. Manure composition updates

Received manure:

In anaerobic lagoon processors, the following nutrient sources are represented in received manure values:

- ManureStream values, as received from the previous processor(s) in the manure management chain
- Precipitation water (kg), calculated in MN.STO.2 (if applicable), is added to the water value in ManureStream

Stored manure:

Below is a summary of updates to ManureStream variables representing the stored manure. Note that the formulas below may be a summarization of multiple steps detailed above, and are intended to provide an overview of what mass losses/gains are reflected in the value of each variable.

Equations in the table below (Calculation column) are in the format of: updated stored manure value = yesterday's stored manure value + today's manure value +/- XYZ. The updated stored manure values reflect the total quantity of manure/nutrients in storage on a single day after accounting for all gains/losses that occurred on that day. Received manure simply refers to the manure being loaded into the manure storage each day.

Table 23: Manure storage variable calculations

Variable	Units	Calculation
water	kg	stored manure water (kg) + received manure water (kg)
total_ammoniacal_nitrogen	kg	max(0, stored manure ammoniacal nitrogen (kg) + received ammoniacal nitrogen (kg) − NH_3N emissions (kg))
nitrogen	kg	stored manure nitrogen (kg) + received manure nitrogen (kg) − NH_3N emissions (kg) − N_2O-N emissions (kg)
phosphorus	kg	stored manure phosphorus (kg) + received manure phosphorus (kg)
potassium	kg	stored manure potassium (kg) + received manure potassium (kg)
ash	kg	stored manure ash (kg) + received manure ash (kg)
degradable_volatile_solids		stored degradable VS (kg) + received degradable VS (kg) − VSd loss (kg)
manure_non_degradable_volatile_-solids	kg	stored manure non-degradable VS (kg) + received manure non-degradable VS (kg) − manure VSnd loss (kg)
bedding_non_degradable_volatile_-solids	kg	stored bedding non-degradable VS (kg) + received bedding non-degradable VS (kg) − bedding VSnd loss (kg)
total_solids	kg	stored total solids (kg) + received total solids (kg) − VSd loss (kg) − VSnd loss (kg)
volume	m^3	stored volume (m^3) + Received volume (m^3) − $\frac{\text{VSd loss (kg)}+\text{VSnd loss (kg)}}{\text{SLURRY\_MANURE\_DENSITY}}$

VIII. Bedded Pack

A. Introduction

Traditional dairy housing systems typically involve concrete flooring and limited bedding, which can lead to cow discomfort, suboptimal health, and laborious manure management. Bedded pack pens aim to address these issues by providing a soft, comfortable, and dry bedding surface for cows, enhancing their comfort and overall well-being.

Bedded packs fall into one of two management methods: a traditional bedded pack, or a compost bedded pack. See UW-Madison Dairyland Initiative Report for further detail. Though the names are often used interchangeably, note here that the primary difference between the two types is the presence or absence of active mixing, and the length of time between complete removal of the accumulated bedding/manure mixture from the pen.

Traditional bedded pack Traditional bedded packs, referred to here as simply “bedded packs”, are typically bedded using straw. Fresh material is added daily without any mixing activity, which causes the manure/bedding mix (pack) to compact and become anaerobic. The pack material is typically removed every 4 to 6 weeks.

Compost Bedded Pack Compost bedded packs are typically bedded with sawdust, shavings, or other fine, dense, absorbent material. Bedding is added daily, accompanied by mixing of the pack material to promote aeration and aerobic decomposition (composting). This composting action leads to the production of heat and microbial activity that promotes breakdown of organic components and reductions in moisture content, resulting in a drier and more stable pack. The pack material is typically removed after several months.

Implementation in RuFaS

The two types of bedded pack are modeled differently in RuFaS as their decomposition conditions and thus emission profiles differ. Bedded packs, where mixing does not occur, promote anaerobic decomposition, whereas compost bedded packs promote aerobic decomposition through regular mixing and aeration of the pack. Sawdust or shavings bedded pens may be used in situations where pens are cleaned more frequently, e.g. weekly, but these pens are not considered to be bedded packs in RuFaS.

Classes

Table 24: List of classes for bedded pack.

Class	Description
CompostBeddedPackBarn(Storage)	bedded_pack.py

Required User Inputs

Table 25: Required inputs for the bedded pack section (refreshed_manure_management.json)

Variable	Definition (units)	Description
Name	none	Unique identifier of the specific bedded pack configuration used.
storage_time_period	days	The number of days that the manure/bedding pack accumulates in the pen before being removed and replaced with fresh bedding.
is_mixed	true/false	A boolean indicator for whether the bedded pack is routinely mixed to intentionally promote composting activity.

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane production potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen
- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids
- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids
- total_solids
- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)
- volume
- methane_production_potential

Expected Outputs

- ManureStream variables representing manure loaded (received) into storage each day, and accumulated manure after accounting for nutrient and mass gains/losses
- storage_methane (kg): Total mass of CH_4 emitted from the bedded pack each day.
- storage_ammonia_N (kg): Total mass of NH_3-N emitted from the bedded pack each day.
- storage_nitrous_oxide_N (kg): Total mass of N2O-N emitted from the bedded pack each day.
- storage_nitrogen_leached (kg): Total mass of N leached from the bedded pack each day. Leached N is assumed to be lost to the environment, and is not captured in runoff that may enter a manure storage.
- carbon_decomposition (kg): the total quantity of manure C lost through microbial degradation of volatile solids.

B. Methodology

Calculate daily methane generation

Description: Calculates the daily mass of methane emitted from the bedded pack based on daily manure VS added to the bedded pack (through animal excretion and bedding addition) and an emission factor based on mixing activity and simulation average temperature (Hanson, Itle, and Edquist, 2024). Here and in other equations, 'daily' denotes the value associated with received manure added to the bedded pack on a specified simulation day. **Note that the quantity of volatile solids utilized in determination of CH_4 includes only manure-excreted volatile solids; bedding volatile solids are excluded.**

Table 26: Methane conversion factor values for bedded pack pens, based on presence or absence of mixing and average annual air temperature.

	Average air temperature (°C)				
Mixing, T/F	≤ 4.6 °C	4.7 - 5.8 °C	5.8 - 13.9 °C	14.0 - 25.1 °C	≥ 25.2 °C
Mixing	0.5	0.5	1.0	1.0	1.5
No mixing	21	26	37	41	74

$$methane\ (kg) = B_0 * MCF * tVS$$

[MN.MET.6]

When:

- B_0 = methane production potential (kg CH_4 per kg manure VS) of manure excreted onto the lot on a specified simulation day
- Constant Value: 0.24 m^3 CH_4/kg VS
- MCF = methane conversion factor (Table 22), based on simulation average ambient temperature

- tVS (kg) = daily mass (kg) of manure-excreted total VS in the bedded pack manure, received from ManureStream(s); note that bedding VS are not included in this value

Calculate carbon decomposition

Description: In addition to microbial processes that occur in anaerobic conditions, which generate primarily CH_4 and CO_2, carbon in the manure/bedding mixture is also degraded through aerobic microbial processes. This process is a function of substrate availability/degradability, temperature, moisture, aeration, and microbial population. This series of calculations is based on the IFSM composting simulation method (Bonifacio, Rotz, and Richard, 2017a; Bonifacio, Rotz, and Richard, 2017b). Simplifications/assumptions that have been made which diverge from the original method are explicitly noted below.

- First, we calculate the maximum decomposition rate per day, and decomposition rate of the slow fraction per day. We use the same equation for both rates, however, the temperature value used in calculating maximum decomposition rate is 60 ℃, versus 30 ℃ in calculating slow fraction degradation. The maximum decomposition rate value is set to 0.04195 and the slow fraction decomposition rate is set to 0.00846, but the equations and set values are shown below for reference.

$$max\_decomp\_rate = EFFECTIVE\_MICROBIAL\_DECOMP\_RATE * (1.066^{DECOMPOSITION\_TEMPERATURE-10} - 1.21^{DECOMPOSITION\_TEMPERATURE-50})$$

[MN.STO.4]

When:

- EFFECTIVE_MICROBIAL_DECOMP_RATE (unitless): The effectiveness of microbial decomposition rate per day, set to 0.00237
- DECOMPOSITION_TEMPERATURE: temperature of the inner compost layer, set to 60 $°C$ (reflective of temperature at which microbial growth, and thus decomposition, is maximized)

$$slow\_decomp\_rate = EFFECTIVE\_MICROBIAL\_DECOMP\_RATE * (1.066^{DEFAULT\_LAYER\_TEMPERATURE-10} - 1.21^{DEFAULT\_LAYER\_TEMPERATURE-50})$$

[MN.STO.5]

When:

- EFFECTIVE_MICROBIAL_DECOMP_RATE (unitless): The effectiveness of microbial decomposition rate per day, set to 0.00237
- DEFAULT_LAYER_TEMPERATURE: temperature of the pack layer, set to 30$°C$ Setting the layer temperature to a constant value is a simplification as manure pack temperature is not modeled dynamically at this time.

- Second, we calculate the carbon decomposition rate per day (calculate_carbon_decomposition_-rate). The value of this parameter is equal to 0.03876, but the equation and set values are included below for reference.

$$C\_decomp\_rate = (max\_decomp\_rate - slow\_decomp\_rate) * e^{FIRST\_ORDER\_DECAYING\_COEFFICIENT*(DEFAULT\_DAYS\_SINCE\_LAST\_MIXING - DEFAULT\_LAG\_TIME)} + slow\_decomp\_rate$$

[MN.STO.6]

When:

- max_decomp_rate and slow_decomp_rate calculated with **[MN.STO.4]**, set to 0.04195 and 0.00846, respectively
- FIRST_ORDER_DECAYING_COEFFICIENT: First-order decaying coefficient constant, set to 0.10
- DEFAULT_DAYS_SINCE_LAST_MIXING: number of days from the start of pack formation or last mixing event, set to 1 by default
- lag: lag time in days to reach maximum decomposition rate, set to 2

- Third, we calculate the anaerobic effect coefficient, related to the effect of the degree of aeration in the manure pack on decomposition. This value is set to 0.9664, but the equation and fixed values are provided below for reference.

$$anaerobic\_effect = \frac{oxygen\_mole\_fraction}{oxygen\_half\_saturation\_constant + oxygen\_mole\_fraction} * \frac{oxygen\_half\_saturation\_constant + oxygen\_ambient\_air_mole\_fraction}{oxygen\_ambient\_air_mole\_fraction} = \frac{0.15}{0.02+0.15} * \frac{0.02+0.21}{0.21} = 0.9664$$

When:

- oxygen_mole_fraction: mole fraction of oxygen in the air within the bedded pack, unitless, set at 0.15. This is a simplification as oxygen content of the manure pack is not currently modeled.
- oxygen_half_saturation_constant: the half-saturation constant, unitless, set at 0.02 by the original publication.
- oxygen_ambient_air_mole_fraction: the mole fraction of oxygen in ambient air, unitless, set at 0.21 (ambient air is approximately 21% oxygen).

- Fourth, we calculate total carbon in the manure/bedding pack available for decomposition. Here we make some assumptions on the carbon content of manure degradable vs. non-degradable volatile solids. Degradable volatile solids, which originate from fecal excretion by animals, are considered to be 50% carbon by weight (Larney, Ellert, and Olson, 2011). Non-degradable volatile solids, which originate primarily from bedding addition, are assumed to contain 35% carbon by weight. The total carbon available is the sum of these two quantities.

$$carbon\_from\_VS_d\ (kg) = degradable\_volatile\_solids * DEFAULT\_CARBON\_FRACTION\_AVAILABLE\_IN\_VSD$$

[MN.STO.7]

When:

- degradable_volatile_solids: The degradable volatile solids (kg) in the daily manure added to the bedded pack.
- DEFAULT_CARBON_FRACTION_AVAILABLE_IN_VSD: the carbon content (%) of manure degradable volatile solids, set to 50% by default.

$$carbon\_from\_VSnd\ (kg) = non\_degradable\_volatile\_solids * DEFAULT\_CARBON\_FRACTION\_AVAILABLE\_IN\_VSND$$

[MN.STO.8]

When:

- nondegradable_volatile_solids: The non-degradable volatile solids (kg) in the daily bedding and manure added to the bedded pack.
- DEFAULT_CARBON_FRACTION_AVAILABLE_IN_VSD: the carbon content (%) of manure degradable volatile solids, set to 35% by default.

$$total\_carbon\ (kg) = carbon\_from\_VSnd + carbon\_from\_VSd$$

[MN.STO.9]

- Finally, we calculate total carbon decomposition in kg/d using the coefficients and values calculated in the steps above (calculate_carbon_decomposition):

$$total\_carbon\_decomposition\ (kg) = (total\_carbon\ *\ C\_decomp\_rate\ *\ DEFAULT\_MOISTURE\_EFFECT\_MICROBIAL\_DECOMP\ *\ anaerobic\_effect)$$

[MN.STO.10]

When:

- total_carbon: total carbon available in manure pack (kg); **[MN.STO.9]**
- C_decomp_rate: carbon decomposition rate per day; **[MN.STO.6]**
- DEFAULT_MOISTURE_EFFECT_MICROBIAL_DECOMP: The effect of moisture on microbial decomposition, set at 0.65. This is a simplification as moisture content of the manure pack is not currently modeled.
- anaerobic_effect: the anaerobic effect coefficient, related to the effect of the degree of aeration in the manure pack on decomposition. Set to 0.9664 by default.

Calculate total VS Loss

_apply_dry_matter_loss

The quantity of total and volatile solids remaining in the accumulated bedded pack must be updated according to estimated CH_4 and C decomposition losses. To do this, we calculate the total loss of VS through CH_4 emission and C decomposition. Loss of mass through CH_4 emissions is assumed to be equal to the mass of CH_4 emitted. Manure volatile solids are assumed to be 50% C, therefore, to determine total mass loss through C decomposition, we divide the mass of C decomposition by 0.50. Importantly, volatile solids destruction is attributed only to manure-excreted volatile solids; bedding volatile solids destruction is not considered at this time.

$$total\_volatile\_solids\_loss(kg) = methane + \frac{total\_carbon\_decomposition}{0.50}$$

[MN.STO.11]

When:

- methane (kg): The daily methane loss, calculated with **[MN.MET.6]**, based on the daily quantity of manure VS added to the bedded pack
- total_carbon_decomposition (kg): quantity of C lost through microbial decomposition (kg), calculated with **[MN.STO.10]**, based on the daily quantity of manure VS added to the bedded pack

Calculate N loss to ammonia

_calculate_cbpb_ammonia_emission

Manure nitrogen being deposited and accumulating in the bedded pack results in NH_3 emissions. Here we utilize an emission factor based on mixing activity (Hanson, Itle, and Edquist, 2024) to estimate total kg of ammonia loss based on the daily manure N deposition by animals.

$$storage\_ammonia\_N\ (kg) = daily\_manure\_N\ (kg) * ammonia\_coefficient$$

[MN.AMM.8]

When:

- daily_maure_N: Daily kg of manure N added to the bedded pack
- ammonia_coefficient: kg of NH_3-N emitted per kg of manure N added per day to the bedded pack, based on the style of management:
  - Bedded pack (no mixing) = AMMONIA_EMISSION_COEFFICIENT_WITH_UNTILLED_-BEDDING (0.25)
  - Compost bedded pack (mixing) = AMMONIA_EMISSION_COEFFICIENT_WITH_TILLED_-BEDDING (0.50)

Calculate N loss to nitrous oxide

_calculate_cbpb_nitrous_oxide_emission

In addition to NH_3-N emissions, manure nitrogen deposition in the bedded pack also results in N2O emissions. Similar to NH_3, we estimate daily N_2O-N loss using an emissions factor from Hanson et al. (2024) based on the type of management used.

$$storage\_nitrous\_oxide\_N\ (kg) = daily\_manure\_N\ (kg) * nitrous\_oxide\_coefficient$$

[MN.NIT.1]

When:

- daily_maure_N: Daily kg of manure N added to the bedded pack
- ammonia_coefficient: kg of NH_3-N emitted per kg of manure N added per day to the bedded pack, based on the style of management:
  - Bedded pack (no mixing) = NITROUS_OXIDE_EMISSION_COEFFICIENT_WITH_UNTILLED_BEDDING (0.01)
  - Compost bedded pack (mixing) = NITROUS_OXIDE_EMISSION_COEFFICIENT_WITH_-TILLED_BEDDING (0.07)

Calculate N loss to leaching

calculate_nitrogen_loss_to_leaching

Manure nitrogen deposited in bedded packs may also be lost to leaching. Leaching of manure N may occur when fecal and urinary N are converted to nitrate in the soil beneath the bedded pack, if the pack is not concrete, lined, or otherwise sealed. Nitrate can then be carried away via water movement through the subsoil. Similar to NH_3 and N_2O, we estimate daily leaching-N loss using an emissions factor from Hanson et al. (2024), which is not influenced by management of the bedded pack (i.e., mixing activity).

$$storage\_leached\_N\ (kg) = daily\_manure\_N\ (kg) * LEACHING\_COEFFICIENT$$

[MN.STO.12]

When:

- daily_maure_N: Daily kg of manure N added to the bedded pack
- LEACHING_COEFFICIENT: kg of N leached per kg of manure N added per day to the open lot; set at 0.035.

C. Received, stored, and emptied outputs

Manure storages in RuFaS report two types of outputs to OutputManager each day: received manure and stored manure.

Received manure

Received manure outputs represent the quantity of manure mass and nutrients added to the manure storage on a single day. No nutrient losses from gas or other emissions/losses are reflected in these output values.

Stored manure

Stored manure outputs represent the accumulated quantity of manure and nutrients present in storage on a single day. These values are the net quantity of mass/nutrients remaining each day after adding received manure values and subtracting any losses to gas emissions or other losses. In bedded pack processors, daily losses include CH_4, NH_3, and N_2O emissions, and N leaching. The order of operations in updating accumulated manure values is:

- Add received manure values to stored manure values
- Calculate gas emissions and total nutrient losses based on stored manure values
- Update stored manure values based on the day's nutrient losses. See the Manure composition update section for specific details on how nutrient gains and losses are accounted for on a daily timestep.

For bedded pack and all other storage processor types, the stored manure values (not received manure) are passed to the next processor in the chain (e.g. another storage, field application, export, etc.) when the storage time interval is complete.

Emptied manure

Manure may be removed from storage via requests made by the Crop and Soil module. The user specifies the days and years for manure removal (i.e. application), as well as the application type (liquid or solid) and quantity of N or P required for each application date within year. Note that these actions are the responsibility of the Crop and Soil module; more information on manure application inputs and methodology can be found in the Crop and Soil module documentation. When manure is removed from storage by the Crop and Soil module, emptied manure outputs report the quantity of manure and nutrients removed on that day, and Manure Stream attributes representing stored manure are updated accordingly to reflect post-removal amounts remaining in storage.

D. Manure composition updates

Received manure

Received manure simply refers to the manure being loaded into the manure storage each day (i.e., deposited in the bedded pack). In bedded pack processors, the following nutrient sources are represented in received manure values:

- ManureStream values, as received from the previous processor(s) in the manure management chain. In bedded pack processors, which are placed first in the manure management chain as there are no intermediary steps between animal excretion and the bedded pack, this ManureStream instance typically represents manure and bedding received directly from the Animal module.
- Daily precipitation volume/mass is **not** currently represented in received bedded pack manure.

Stored manure

Below is a summary of updates to ManureStream variables representing the stored manure. Note that the formulas below may be a summarization of multiple steps detailed above, and are intended to provide an overview of what mass losses/gains are reflected in the value of each variable.

Equations in the table below (Calculation column) are in the format of: updated stored manure value = yesterday's stored manure value + today's manure value +/- XYZ. The updated stored manure values reflect the total quantity of manure/nutrients in storage on a single day after accounting for all gains/losses that occurred on that day. Received manure simply refers to the manure being loaded into the manure storage each day.

Table 27: Calculated manure storage variables and their units

Variable	Units	Calculation
water	kg	Stored manure water (kg) + received manure water (kg)

Continued on next page

Variable	Units	Calculation
total_ammoniacal_nitrogen	kg	$\max(0, \text{stored manure ammoniacal N} + \text{received ammoniacal N} - \text{NH}_3\text{N emissions})$
nitrogen	kg	Stored manure N + received manure N − NH_3-N − N_2O-N emissions
phosphorus	kg	Stored manure P + received manure P
potassium	kg	Stored manure K + received manure K
ash	kg	Stored manure ash + received manure ash
degradable_volatile_solids		stored degradable VS (kg)+received degradable VS (kg)− VSd loss (kg)
manure_non_degradable_volatile_-solids	kg	stored manure non-degradable VS (kg) + received manure non-degradable VS (kg) − VSnd loss (kg)
bedding_non_degradable_volatile_-solids	kg	stored bedding non-degradable VS (kg) + received bedding non-degradable VS (kg)
total_solids	kg	Stored TS + received TS − VS loss
volume	m^3	$\text{Stored volume} + \text{received volume} - \left(\frac{\text{VS loss}}{\text{SOLID\_MANURE\_DENSITY}}\right)$

IX. Open Lot

A. Introduction

Open or dry lot dairy systems are systems in which cows are housed outdoors in earthen or concrete pens. The open lot surface typically consists of natural or compacted soil covered with accumulated dry manure. Open lot pens often contain some form of shelter to provide shade and protection from harsh weather conditions, under which bedding may be applied, especially during winter. In addition to the dirt/manure pack lot, these systems also typically have a segregated feed feeding area, which may include a concrete apron that cows stand and deposit manure upon while eating. Manure deposited in this area may be managed differently from manure deposited on the lot surface.

To manage the manure on the lot surface, farmers commonly harrow the surface to spread out, break up, and mix fresh manure into the existing dry pack, to facilitate drying and create a standing and lying surface mainly consisting of dry manure solids. Harrowing is usually performed daily, which results in mixing of the soil/dry manure pack and fresh manure and urine. When cows deposit urine and manure on the lot surface, the dry climate and low humidity typical to regions where these systems are common facilitate rapid drying and ammonia (NH_3) volatilization from urine. Harrowing activity also promotes volatilization of NH_3 through mixing of fresh urine and manure. Though NH_3 losses may be high, the harrowing/spreading action also serves to create a lighter, more aerated pack that emits less methane (CH_4) than other manure storage methods, such as slurry or liquid manure storage; however, these same aerobic conditions can also lead to greater nitrous oxide (N_2O) emissions from the lot surface.

Manure is allowed to accumulate on the lot surface for several weeks to months, and may be either spread flat or partially piled in the pen during this period. Lots are cleaned out typically via scraping with large equipment 1-2 times per year. Manure removed from lots may be immediately field applied, stacked/piled until field application, composted, or managed in other ways. Of note is that, while the majority of manure on open lot dairies may be managed as a solid, open lot dairies with milking animals will also utilize a slurry or liquid manure storage to store, at a minimum, milking parlor waste. Feed apron waste, lot runoff, or other waste may also be stored as a liquid.

Implementation in RuFaS

In the Open Lot processor, we receive manure information from the animal module and manure management information from user inputs. Given weather data, we then calculate daily nutrient losses through CH_4, N_2O, and NH_3 emissions, as well as N leaching and C decomposition. We update the composition of the accumulated manure mix each day as manure and urine are added and nutrients are lost through the processes mentioned.

Important assumptions:

- Harrowing is assumed to be performed daily. At this time, differences in lot manure harrowing or piling frequency are not reflected in calculation of emissions/losses.
- Gas emissions in this processor are based on annual emissions factors that are extrapolated to a daily timestep, meaning that calculations are made based on daily excretion of manure VS or N, rather than accumulated quantities of the nutrients. With this, for N and NH_3 emissions for example, manure is assumed to lose all estimated NH_3-N on the day it is excreted. Manure N accumulation in lot manure is tracked, but accumulated manure N does not influence NH_3 emissions.

Classes

Table 28: Classes available in open lots.

	Description
Storage(OpenLot)	open_lot.py.

Required User Inputs

Table 29: Required inputs for the open lots section

Variable	Units	Description
storage_time_period	days	The interval in days that the open lot pen is cleaned out, i.e., the majority of manure, either already in piles or spread across the lot surface, is scraped or otherwise collected from the lot and removed from the pen.

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen
- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids
- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids
- total_solids
- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)
- volume
- methane_production_potential

Expected Outputs

- ManureStream variables representing manure loaded (received) into storage each day, and accumulated manure after accounting for nutrient and mass gains/losses
- storage_methane (kg): Total mass of CH_4 emitted from the open lot manure each day.
- storage_ammonia_N (kg): Total mass of NH_3-N emitted from the open lot manure each day.
- storage_nitrous_oxide_N (kg): Total mass of N2O-N emitted from the open lot manure each day.
- storage_nitrogen_leached (kg): Total mass of N leached from the open lot manure each day. Leached N is assumed to be lost to the environment, and is not captured in lot runoff that may enter a manure storage.
- carbon_decomposition (kg): the total quantity of manure C lost through microbial degradation of volatile solids.

B. Methodology

Calculate methane conversion factor

calculate_ifsm_methane_emission

Calculates the methane conversion factor for open lot manure given the ambient air temperature using an equation from IFSM (Rotz, Corson, et al., 2023). This method is an adaptation of the IPCC, 2006 tier 2 approach. Note that the MCF cannot be lower than 0.

$$MCF = max(0, \frac{0.0625 * T - 0.25}{100})$$

[MN.MET.5]

When:

- T = ambient air temperature, 0 to 30$°C$

Calculate daily methane generation

calculate_ifsm_methane_emission

Calculates the daily mass of methane emitted from the open lot based on daily manure volatile solids (VS) added to lot manure (through animal excretion and bedding addition) and the emission factor calculated in **[MN.MET.5]**. **Note that the quantity of volatile solids utilized in determination of CH_4 includes only manure-excreted volatile solids; bedding volatile solids are excluded.** Here and in other equations, ‘daily’ denotes the value associated with received manure added to the open lot on a specified simulation day.

$$methane\ (kg) = B_0 * MCF * tVS * METHANE\_FACTOR$$

[MN.MET.6]

When:

- B_0 = methane production potential (kg CH_4 per kg manure VS) of manure excreted onto the lot on a specified simulation day
- MCF = methane conversion factor calculated in **[MN.MET.5]**, based on daily ambient temperature

- tVS (kg) = daily mass (kg) of manure-excreted volatile solids in the open lot pen, received from ManureStream(s); bedding volatile solids are not included in this value
- METHANE_FACTOR = 0.67; unit conversion factor for CH_4 volume to mass (kg).

Calculate carbon decomposition

calculate_carbon_decomposition

In addition to microbial processes that occur in anaerobic conditions, which generate primarily CH_4 and CO_2, open lot manure C is also degraded through aerobic microbial processes. This process is a function of substrate availability/degradability, temperature, moisture, aeration, and microbial population. This series of calculations is based on the IFSM composting simulation method, which is described in detail in Bonifacio, Rotz, and Richard, 2017a and Bonifacio, Rotz, and Richard, 2017b. Simplifications/assumptions that have been made which diverge from the original method are explicitly noted below.

- First, we calculate the maximum decomposition rate per day, and decomposition rate of the slow fraction per day. We use the same equation for both rates, however, the temperature value used in calculating maximum decomposition rate is $60°C$, versus $30°C$ in calculating slow fraction degradation. The maximum decomposition rate value is set to 0.04195 and the slow fraction decomposition rate is set to 0.00846, but the equations and set values are shown below for reference.

$$max\_decomp\_rate = EFFECTIVE\_MICROBIAL\_DECOMP\_RATE * 1.066^{(DECOMPOSITION\_TEMPERATURE-10)-1.21^{(}DECOMPOSITION\_TEMPERATURE-50)}$$

[MN.STO.4]

When:

- EFFECTIVE_MICROBIAL_DECOMP_RATE (unitless): The effectiveness of microbial decomposition rate per day, set to 0.00237
- DECOMPOSITION_TEMPERATURE: temperature of the inner compost layer, set to $60°C$ (reflective of temperature at which microbial growth, and thus decomposition, is maximized)

$$slow\_decomp\_rate = EFFECTIVE\_MICROBIAL\_DECOMP\_RATE * 1.066^{(DEFAULT\_LAYER\_TEMPERATURE-10)} - 1.21^{(DEFAULT\_LAYER\_TEMPERATURE-50)}$$

[MN.STO.5]

When:

- EFFECTIVE_MICROBIAL_DECOMP_RATE (unitless): The effectiveness of microbial decomposition rate per day, set to 0.00237
- DECOMPOSITION_TEMPERATURE: temperature of the inner compost layer, set to $30°C$. Setting the layer temperature to a constant value is a simplification as manure pack temperature is not modeled dynamically at this time.

- Second, we calculate the carbon decomposition rate per day (calculate_carbon_decomposition_-rate). The value of this parameter is equal to 0.03876, but the equation and set values are included below for reference.

$$C\_decomp\_rate = (max\_decomp\_rate - slow\_decomp\_rate) * e^{FIRST\_ORDER\_DECAYING\_COEFFICIENT*(DEFAULT_DAYS\_SINCE\_LAST\_MIXING - DEFAULT\_LAG\_TIME)} + slow\_decomp_rate$$

[MN.STO.6]

When:

- max_decomp_rate and slow_decomp_rate calculated with **[MN.STO.4]**, set to 0.04195 and 0.00846, respectively
- FIRST_ORDER_DECAYING_COEFFICIENT: First-order decaying coefficient constant, set to 0.10
- DEFAULT_DAYS_SINCE_LAST_MIXING: number of days from the start of pack formation or last mixing event, set to 1 by default
- lag: lag time in days to reach maximum decomposition rate, set to 2

- Third, we calculate the anaerobic effect coefficient, related to the effect of the degree of aeration in the manure pack on decomposition. This value is set to 0.9664, but the equation and fixed values are provided below for reference.

$$anaerobic\_effect = \frac{oxygen\_mole\_fraction}{(oxygen\_half\_saturation\_constant + oxygen\_mole\_fraction)} * \frac{(oxygen\_half\_saturation\_constant + oxygen\_ambient\_air\_mole\_fraction)}{oxygen\_ambient\_air\_mole\_fraction} = \frac{0.15}{(0.02+0.15)} * \frac{(0.02+0.21)}{0.21} = 0.9664$$

When:

- oxygen_mole_fraction: mole fraction of oxygen in the air within the lot manure pack, unitless, set at 0.15. This is a simplification as oxygen content of the manure pack is not currently modeled.
- oxygen_half_saturation_constant: the half-saturation constant, unitless, set at 0.02 by the original publication.
- oxygen_ambient_air_mole_fraction: the mole fraction of oxygen in ambient air, unitless, set at 0.21 (ambient air is approximately 21% oxygen).

- Fourth, we calculate total carbon in the manure/bedding pack available for decomposition. Here we make some assumptions on the carbon content of manure degradable vs. non-degradable volatile solids. Degradable volatile solids, which originate from fecal excretion by animals, are considered to be 50% carbon by weight (Larney, Ellert, and Olson, 2011). Non-degradable volatile solids, which originate primarily from bedding addition, are assumed to contain 35% carbon by weight. The total carbon available is the sum of these two quantities.

$$carbon\_from\_VSd\ (kg) = degradable\_volatile\_solids * DEFAULT\_CARBON\_FRACTION\_AVAILABLE\_IN\_VSD$$

[MN.STO.7]

When:

- degradable_volatile_solids: The degradable volatile solids (kg) in the daily manure added to the open lot.
- DEFAULT_CARBON_FRACTION_AVAILABLE_IN_VSD: the carbon content (%) of manure degradable volatile solids, set to 50% by default.

$$carbon\_from\_VSnd\ (kg) = non\_degradable\_volatile\_solids * DEFAULT\_CARBON_FRACTION\_AVAILABLE\_IN\_VSND$$

[MN.STO.8]

When:

- non_degradable_volatile_solids: The non-degradable volatile solids (kg) in the daily manure added to the open lot.
- DEFAULT_CARBON_FRACTION_AVAILABLE_IN_VSD: the carbon content (%) of manure non-degradable volatile solids, set to 35% by default.

$$total\_carbon\ (kg) = carbon\_from\_VS_{nd} + carbon\_from\_VS_{d}$$

[MN.STO.9]

- Finally, we calculate total carbon decomposition in kg/d using the coefficients and values calculated in the steps above (_apply_dry_matter_loss):

$$total\_carbon\_decomposition\ (kg) = (total\_carbon * C\_decomp\_rate * DEFAULT\_MOISTURE\_EFFECT\_MICROBIAL\_DECOMP * anaerobic\_effect)$$

[MN.STO.10]

When:

- total_carbon: total carbon available in manure pack (kg); **[MN.STO.9]**
- C_decomp_rate: carbon decomposition rate per day; **[MN.STO.10]**
- DEFAULT_MOISTURE_EFFECT_MICROBIAL_DECOMP: The effect of moisture on microbial decomposition, set at 0.65. This is a simplification as moisture content of the manure pack is not currently modeled.
- anaerobic_effect: the anaerobic effect coefficient, related to the effect of the degree of aeration in the manure pack on decomposition. Set to 0.9664 by default.

Calculate total VS Loss

The quantity of total and volatile solids remaining in the accumulated bedded pack must be updated according to estimated CH_4 and C decomposition losses. To do this, we calculate the total loss of VS through CH_4 emission and C decomposition. Loss of mass through CH_4 emissions is assumed to be equal to the mass of CH_4 emitted. Manure volatile solids are assumed to be 50% C, therefore, to determine total mass loss through C decomposition, we divide the mass of C decomposition by 0.50. Importantly, volatile solids destruction is attributed only to manure-excreted volatile solids; bedding volatile solids destruction is not considered at this time.

$$total\_volatile\_solids\_loss(kg) = methane + \frac{total\_carbon\_decomposition}{0.50}$$

[MN.STO.11]

When:

- methane (kg): The daily methane loss, calculated with **[MN.MET.6]**, based on the daily quantity of manure VS added to the open lot
- total_carbon_decomposition (kg): quantity of C lost through microbial decomposition (kg), calculated with **[MN.STO.10]**, based on the daily quantity of manure VS added to the open lot

Calculate N loss to ammonia

_calculate_cbpb_ammonia_emission

Manure nitrogen being deposited and accumulating in the bedded pack results in NH_3 emissions. Here we utilize an emission factor based on mixing activity (Hanson, Itle, and Edquist, 2024) to estimate total kg of ammonia loss based on the daily manure N deposition by animals.

$$storage\_ammonia\_N\ (kg) = daily\_manure\_N * ammonia\_coefficient$$

[MN.AMM.8]

When:

- daily_maure_N: Daily kg of manure N added to the open lot
- AMMONIA_EMISSION_COEFFICIENT_IN_OPEN_LOTS: kg of NH_3-N emitted per kg of manure N added per day to the open lot; set at 0.36.

Calculate N loss to nitrous oxide

_calculate_cbpb_nitrous_oxide_emission

In addition to NH_3-N emissions, manure nitrogen deposition in the bedded pack also results in N_2O emissions. Similar to NH_3, we estimate daily N2O-N loss using an emissions factor from Hanson, Itle, and Edquist, 2024 based on the type of management used.

$$storage\_nitrous\_oxide\_N\ (kg) = daily\_manure\_N * nitrous\_oxide\_coefficient$$

[MN.NIT.1]

When:

- daily_maure_N: Daily kg of manure N added to the open lot
- NITROUS_OXIDE_COEFFICIENT_IN_OPEN_LOTS: kg of N_2-O-N emitted per kg of manure N added per day to the open lot; set at 0.02.

Calculate N loss to leaching

calculate_nitrogen_loss_to_leaching

Manure nitrogen deposited in bedded packs may also be lost to leaching. Leaching of manure N may occur when fecal and urinary N are converted to nitrate in the soil beneath the bedded pack, if the pack is not concrete, lined, or otherwise sealed. Nitrate can then be carried away via water movement through the subsoil. Similar to NH_3 and N_2O, we estimate daily leaching-N loss using an emissions factor from Hanson, Itle, and Edquist, 2024, which is not influenced by management of the bedded pack (i.e., mixing activity).

$$storage\_leached\_N\ (kg) = daily\_manure\_N\ (kg) * LEACHING\_COEFFICIENT$$

[MN.STO.12]

When:

- daily_maure_N: Daily kg of manure N added to the open lot
- LEACHING_COEFFICIENT: kg of N leached per kg of manure N added per day to the open lot; set at 0.035.

C. Received, stored, and emptied outputs

Manure storages in RuFaS report two types of outputs to OutputManager each day: received manure and stored manure.

Received manure

Received manure outputs represent the quantity of manure mass and nutrients added to the manure storage on a single day. No nutrient losses from gas or other emissions/losses are reflected in these output values.

Stored manure

Stored manure outputs represent the accumulated quantity of manure and nutrients present in storage on a single day. These values are the net quantity of mass/nutrients remaining each day after adding received manure values and subtracting any losses to gas emissions or other losses. In open lot processors, daily losses include CH_4, NH_3, and N_2O emissions, and N leaching. The order of operations in updating accumulated manure values is:

- Add received manure values to stored manure values
- Calculate gas emissions and total nutrient losses based on stored manure values
- Update stored manure values based on the day's nutrient losses. See the Manure composition update section for specific details on how nutrient gains and losses are accounted for on a daily timestep.

For open lot and all other storage processor types, the stored manure values (not received manure) are passed to the next processor in the chain (e.g. another storage, field application, export, etc.) when the storage time interval is complete.

Emptied manure

Manure may be removed from storage via requests made by the Crop and Soil module. The user specifies the days and years for manure removal (i.e. application), as well as the application type (liquid or solid) and quantity of N or P required for each application date within year. Note that these actions are the responsibility of the Crop and Soil module; more information on manure application inputs and methodology can be found in the Crop and Soil module documentation. When manure is removed from storage by the Crop and Soil module, emptied manure outputs report the quantity of manure and nutrients removed on that day, and Manure Stream attributes representing stored manure are updated accordingly to reflect post-removal amounts remaining in storage.

D. Manure composition updates

Received manure

Received manure simply refers to the manure being loaded into the manure storage each day (i.e., deposited on the lot). In open lot processors, the following nutrient sources are represented in received manure values:

- ManureStream values, as received from the previous processor(s) in the manure management chain. In open lot processors, which are placed first in the manure management chain as there are no intermediary steps between animal excretion and the open lot, this ManureStream instance typically represents manure and bedding received directly from the Animal module.
- Daily precipitation volume/mass is **not** currently represented in received open lot manure.

Stored manure

Below is a summary of updates to ManureStream variables representing the stored manure. Note that the formulas below may be a summarization of multiple steps detailed above, and are intended to provide an overview of what mass losses/gains are reflected in the value of each variable.

Equations in the table below (Calculation column) are in the format of: updated stored manure value = yesterday's stored manure value + today's manure value +/- XYZ. The updated stored manure values reflect the total quantity of manure/nutrients in storage on a single day after accounting for all gains/losses that occurred on that day. Received manure simply refers to the manure being loaded into the manure storage each day.

Table 30: Calculated manure storage variables and their units

Variable	Units	Calculation
water	kg	Stored manure water (kg) + received manure water (kg)
total_ammoniacal_nitrogen	kg	max(0, stored manure ammoniacal N + received ammoniacal N − NH_3N emissions)
nitrogen	kg	Stored manure N + received manure N − NH_3-N − N_2O-N emissions
phosphorus	kg	Stored manure P + received manure P
potassium	kg	Stored manure K + received manure K
ash	kg	Stored manure ash + received manure ash
degradable_volatile_solids		stored degradable VS (kg)+received degradable VS (kg)− VSd loss (kg)

Continued on next page

Variable	Units	Calculation
manure_non_degradable_volatile_-solids	kg	stored manure non-degradable VS (kg) + received manure non-degradable VS (kg) − VSnd loss (kg)
bedding_non_degradable_volatile_-solids	kg	stored bedding non-degradable VS (kg) + received bedding non-degradable VS (kg)
total_solids	kg	Stored TS + received TS − VS loss
volume	m^3	Stored volume + received volume − $\left(\frac{\text{VS loss}}{\text{SOLID\_MANURE\_DENSITY}}\right)$

[1] The "Max(0,)" notation prevents the ammoniacal N value from becoming negative. This is especially important early in a simulation when accumulated manure quantities that are very small compared to the fixed surface area value can lead to high ammonia emissions.

[2] degradable_volatile_solids_frac = received manure degradable_volatile_solids / received manure total_volatile_solids

X. Composting

A. Introduction

Composting is a method of treating manure in which solid manure, generally with bedding or other carbon-rich material added, is purposefully managed in order to promote microbial decomposition. This naturally occurring decomposition process results in losses of both water and organic matter, resulting in a smaller volume and mass of manure the farm is required to store and handle. Composting also helps to stabilize manure by killing pathogens and weed seeds via heat generated from the decomposition process, and reduced odors. Finished compost is a useful end-product (like all well-managed manure) that can be land-applied, sold, or otherwise exported.
To promote optimal conditions for microbial decomposition, compost requires airflow through the pile to prevent anaerobic conditions from forming. While turning is the traditional way to achieve this, several methods to accomplish the necessary aeration of compost exist on farms today.

Implementation in RuFaS The following composting methods are represented in RuFaS:

- Static pile: material to be composted is piled and not turned/moved throughout the composting process. Aeration is maintained using either bulking agents to create airspace in the pile, or through forced air introduction via pipes.
- Intensive windrow: material to be composted is piled into long rows that are regularly turned to redistribute moisture, nutrients, and air through the pile.
- Passive windrow: material to be composted is piled into long rows, but is not regularly turned, and instead relies on passive air diffusion to aerate the pile.
- In-vessel: material to be composted is loaded into a large vessel, often a drum, which is mechanically rotated to promote uniform air circulation and heating.

At this time, turning frequency, pile composition, composting duration, C:N ratio, and other factors do not directly affect emissions or nutrient loss calculations. Only the general composting method and the basic composition (kg of N, volatile solids, etc.) of material added to the pile is factored into this submodule's calculations at this time.

Classes

Table 31: Classes available in composting.

	Description
Storage(Composting)	composting.py.

Required User Inputs

Table 32: Required inputs for the composting section (refreshed_manure_management.json)

Variable	Definition (units)	Description
Name	none	Unique identifier of the specific handler configuration used.
Storage_time_period	days	The number of days that manure is stored between emptying events. At the end of this interval, the manure storage is emptied completely.
composting_type	static pile, passive windrow, intensive windrow, or in-vessel	The type of method used for composting

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane production potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen
- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids
- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids
- total_solids
- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)
- volume
- methane_production_potential

Expected Outputs

- ManureStream variables representing manure loaded (received) into storage each day, and accumulated manure after accounting for nutrient and mass gains/losses
- storage_methane (kg): Total mass of CH_4 emitted from the compost each day.
- storage_ammonia_N (kg): Total mass of NH_3-N emitted from the compost each day.

- storage_nitrous_oxide_N (kg): Total mass of N_2O-N emitted from the compost each day.
- storage_nitrogen_leached (kg): Total mass of N leached from the compost each day. Leached N is assumed to be lost to the environment, and is not captured in runoff that may enter a manure storage.
- carbon_decomposition (kg): the total quantity of manure C lost through microbial degradation of volatile solids.

B. Methodology

Calculate daily methane generation

_calculate_composting_methane_emissions

Calculates the daily mass of methane emitted from the compost based on manure volatile solids (VS) added to compost (from animal excretion and bedding addition) and an emission factor based on composting method and simulation average temperature described in Table 1 (Hanson, Itle, and Edquist, 2024). **Note that the quantity of volatile solids utilized in determination of CH_4 includes only manure-excreted volatile solids; bedding volatile solids are excluded.** Here and in other equations, 'daily' denotes the value associated with received manure added to the open lot on a specified simulation day.

Table 33: Methane conversion factor values for compost pens, based on composting method and average annual air temperature.

	Average air temperature (°C)		
Composting method	0 - 10 °C	10 - 18 °C	>18 °C
In-vessel	0.5	0.5	0.5
Static pile	1.0	2.0	2.5
Intensive windrow	0.5	1.0	1.5
Passive windrow	1.0	2.0	2.5

$$methane\ (kg) = B_o * MCF * tVS$$

[MN.MET.6]

When:

- B_0 = methane production potential (kg CH_4 per kg manure VS) of manure excreted into the bedded pack on a specified simulation day
- MCF = methane conversion factor, based on annual ambient temperature
- tVS (kg) = daily mass (kg) of manure-excreted volatile solids in the open lot pen, received from ManureStream(s); bedding volatile solids are not included in this value
- METHANE_FACTOR = 0.67; unit conversion factor for CH_4 volume to mass (kg).

Calculate carbon decomposition

calculate_carbon_decomposition

Carbon in the compost material is degraded through primarily aerobic microbial processes. This process is a function of substrate availability/degradability, temperature, moisture, aeration, and microbial population. This series of calculations is based on the IFSM composting simulation method, which is described in detail in Bonifacio, Rotz, and Richard, 2017a and Bonifacio, Rotz, and Richard, 2017b. Simplifications/assumptions that have been made which diverge from the original method are explicitly noted below.

- First, we calculate the maximum decomposition rate per day, and decomposition rate of the slow fraction per day. We use the same equation for both rates, however, the temperature value used in calculating maximum decomposition rate is $60°C$, versus the actual ambient air temperature in calculating slow fraction degradation. The maximum decomposition rate value is set to 0.04195, but the equation is shown below for reference.

$$max\_decomp\_rate = EFFECTIVE\_MICROBIAL\_DECOMP\_RATE * (1.066^{(DECOMPOSITION\_TEMPERATURE-10)} - 1.21^{(DECOMPOSITION\_TEMPERATURE-50)})$$

[MN.STO.4]

  - EFFECTIVE_MICROBIAL_DECOMP_RATE (unitless): The effectiveness of microbial decomposition rate per day, set to 0.00237
  - DECOMPOSITION_TEMPERATURE: temperature of the inner compost layer, set to 60∘C (reflective of temperature at which microbial growth, and thus decomposition, is maximized)

$$slow\_decomp\_rate = EFFECTIVE\_MICROBIAL\_DECOMP\_RATE * 1.066^{(daily\_temperature-10)} - 1.21^{(daily\_temperature-50)}$$

[MN.STO.5]

  - EFFECTIVE_MICROBIAL_DECOMP_RATE (unitless): The effectiveness of microbial decomposition rate per day, set to 0.00237
  - daily_temperature: average ambient air temperature on a single day (∘C).

- Second, we calculate the carbon decomposition rate per day (calculate_carbon_decomposition_-rate). The value of this parameter is equal to 0.03876, but the equation and set values are included below for reference.

$$C\_decomp\_rate = (max\_decomp\_rate - slow\_decomp\_rate) * e^{(FIRST\_ORDER\_DECAYING\_COEFFICIENT*(DEFAULT\_DAYS\_SINCE\_LAST\_MIXING-DEFAULT\_LAG\_TIME))} + slow\_decomp\_rate$$

[MN.STO.6]

  - max_decomp_rate and slow_decomp_rate calculated with **[MN.STO.4]** and **[MN.STO.5]**

- FIRST_ORDER_DECAYING_COEFFICIENT: First-order decaying coefficient constant, set to 0.10
- DEFAULT_DAYS_SINCE_LAST_MIXING: number of days from the start of composting or last turning/mixing event, set to 1 (i.e., assuming daily mixing)
- lag: lag time in days to reach maximum decomposition rate, set to 2

- Third, we calculate the anaerobic effect coefficient, related to the effect of the degree of aeration in the manure pack on decomposition. This value is set to 0.9664, but the equation and fixed values are provided below for reference.

$$anaerobic\_effect = (\frac{oxygen\_mole\_fraction}{(oxygen\_half\_saturation\_constant+oxygen\_mole\_fraction)}) * (\frac{(oxygen\_half\_saturation\_constant+oxygen\_ambient\_air\_mole\_fraction)}{oxygen\_ambient\_air\_mole\_fraction}) = (\frac{0.15}{(0.02+0.15)}) * (\frac{(0.02+0.21)}{0.21}) = 0.9664$$

  - oxygen_mole_fraction: mole fraction of oxygen in the air within the windrow, unitless, set at 0.15. This is a simplification as oxygen content of the manure pack is not currently modeled.
  - oxygen_half_saturation_constant: the half-saturation constant, unitless, set at 0.02 by the original publication.
  - oxygen_ambient_air_mole_fraction: the mole fraction of oxygen in ambient air, unitless, set at 0.21 (ambient air is approximately 21% oxygen).

- Fourth, we calculate total carbon in the compost material available for decomposition. Here we make some assumptions on the carbon content of manure degradable vs. non-degradable volatile solids. Degradable volatile solids, which originate from fecal excretion by animals, are considered to be 50% carbon by weight (Larney, Ellert, and Olson, 2011). Non-degradable volatile solids, which originate primarily from bedding addition, are assumed to contain 35% carbon by weight. The total carbon available is the sum of these two quantities.

$$carbon\_from\_VSd\ (kg) = degradable\_volatile\_solids * DEFAULT\_CARBON\_FRACTION\_AVAILABLE\_IN\_VSD$$

[MN.STO.7]

  - degradable_volatile_solids: The degradable volatile solids (kg) in the daily manure added to the compost.
  - DEFAULT_CARBON_FRACTION_AVAILABLE_IN_VSD: the carbon content (%) of manure degradable volatile solids, set to 50% by default.

$$carbon\_from\_VSnd\ (kg) = non\_degradable\_volatile\_solids * DEFAULT\_CARBON\_FRACTION\_AVAILABLE\_IN\_VSND$$

[MN.STO.8]

- non_degradable_volatile_solids: The non-degradable volatile solids (kg) in the daily bedding and manure added to the compost.
- DEFAULT_CARBON_AVAILABLE_IN_VSND: the carbon content (%) of manure non-degradable volatile solids, set to 35% by default.

$$total\_carbon\ (kg) = carbon\_from\_VSnd + carbon\_from\_VSd$$

[MN.STO.9]

- Finally, we calculate total carbon decomposition in kg/d using the coefficients and values calculated in the steps above (calculate_carbon_decomposition):

$$total\_carbon\_decomposition\ (kg) = (total\_carbon * C\_decomp\_rate * DEFAULT\_MOISTURE\_EFFECT\_MICROBIAL\_DECOMP * anaerobic\_effect)$$

[MN.STO.10

- total_carbon: total carbon available in compost (kg); **[MN.STO.9]**
- C_decomp_rate: carbon decomposition rate per day; **[MN.STO.6]**
- DEFAULT_MOISTURE_EFFECT_MICROBIAL_DECOMP: The effect of moisture on microbial decomposition, set at 0.65. This is a simplification as moisture content of the compost is not currently modeled.
- anaerobic_effect: the anaerobic effect coefficient, related to the effect of the degree of aeration in the compost on decomposition. Set to 0.9664 by default.

Calculate total VS loss
_apply_dry_matter_loss

The quantity of total and volatile solids remaining in the compost each day must be updated according to estimated CH_4 and C decomposition losses. To do this, we calculate the total daily loss of VS through CH_4 emission and C decomposition. Loss of mass through CH_4 emissions is assumed to be equal to the mass of CH_4 emitted. Manure volatile solids are assumed to be 50% C, therefore, to determine total mass loss through C decomposition, we divide the mass of C decomposition by 0.50.

$$total\_volatile\_solids\_loss\ (kg) = methane + (\frac{total\_carbon\_decomposition}{0.50})$$

[MN.STO.11

- methane (kg): The daily methane loss, calculated with **[MN.MET.6]**, based on the daily quantity of manure VS added to the compost
- total_carbon_decomposition (kg): quantity of C lost through microbial decomposition (kg), calculated with **[MN.STO.10]**, based on the daily quantity of manure VS added to the compost

Calculate N loss to ammonia
_calculate_composting_ammonia_emissions

Description: Manure nitrogen being added to and accumulating in the compost results in NH_3 emissions. Here we utilize daily manure N addition to compost and an emission factor based on composting method (Hanson, Itle, and Edquist, 2024) to estimate total kg of NH_3-N loss.

Table 34: NH_3-N emission factors for compost by composting method.

Composting method	kg NH_3-N emitted per kg of N added to compost
In-vessel	0.45
Static pile	0.50
Intensive windrow	0.50
Passive windrow	0.45

$$storage\_ammonia\_N\ (kg) = daily\_manure\_N\ (kg) * ammonia\_coefficient$$

[MN.AMM.8

- daily_maure_N: Daily kg of manure N added to compost
- ammonia_coefficient: kg of NH_3-N emitted per kg of manure N added per day to compost, based on the composting method

Calculate N loss to nitrous oxide
_calculate_nitrous_oxide_emissions

Description: In addition to NH_3-N emissions, nitrogen added to and accumulating in the compost also results in N_2O emissions. Similar to NH_3, here we utilize daily manure N addition to compost and an emission factor based on composting method (Hanson, Itle, and Edquist, 2024) to estimate total kg of $N_2O - N$ loss.

Composting method	kg NH_3-N emitted per kg of N added to compost
In-vessel	0.006
Static pile	0.010
Intensive windrow	0.005
Passive windrow	0.005

$$storage\_nitrous\_oxide\_N\ (kg) = daily\_manure\_N\ (kg) * nitrous\_oxide\_coefficient$$

[MN.NIT.1

- daily_maure_N: daily kg of manure N added to the compost
- nitrous_oxide_coefficient: kg of $N_2O - N$ emitted per kg of manure N added per day to compost, based on the composting method

Calculate N loss to leaching
calculate_nitrogen_loss_to_leaching

Description: Nitrogen in compost may also be lost to leaching. Leaching of compost N may occur when fecal and urinary N are converted to nitrate in the soil beneath the compost, if the compost is not contained in a vessel. Nitrate can then be carried away via water movement through the subsoil. Similar to NH_3 and N_2O, we use daily manure N addition to compost and an emission factor based on composting method (Hanson, Itle, and Edquist, 2024) to estimate total kg of leaching N loss.

Table 35: N_2O-N emission factors for compost by composting method.

Composting method	**kg NH_3-N emitted per kg of N added to compost**
In-vessel	0.00
Static pile	0.06
Intensive windrow	0.06
Passive windrow	0.04

$$storage\_leached\_N\ (kg) = daily\_manure\_N\ (kg) * leaching\_coefficient$$

[MN.STO.12]

- daily_maure_N: daily kg of manure N added to the bedded pack
- leaching_coefficient: kg of N leached per kg of manure N added per day to compost, based on the composting method

C. Received, stored, and emptied outputs

Received manure

Received manure outputs represent the quantity of manure mass and nutrients added to the manure storage on a single day. No nutrient losses from gas or other emissions/losses are reflected in these output values.

Stored manure

Stored manure outputs represent the accumulated quantity of manure and nutrients present in storage on a single day. These values are the net quantity of mass/nutrients remaining each day after adding received manure values and subtracting any losses to gas emissions or other losses. In composting processors, daily losses include CH_4, NH_3, N leaching, and N2O emissions. The order of operations in updating accumulated manure values is:

1. Add received manure values to stored manure values
2. Calculate gas emissions and total nutrient losses based on received manure values
3. Update stored manure values based on the day's nutrient losses. See the Manure composition update section for specific details on how nutrient gains and losses are accounted for on a daily timestep.

For composting and all other storage processor types, the stored manure values (not received manure) are passed to the next processor in the chain (e.g. another storage, field application, export, etc.) when the storage time interval is complete.

Emptied manure

Manure may be removed from storage via requests made by the Crop and Soil module. The user specifies the days and years for manure removal (i.e. application), as well as the application type (liquid or solid) and quantity of N or P required for each application date within year. Note that these actions are the responsibility of the Crop and Soil module; more information on manure application inputs and methodology can be found in the Crop and Soil module documentation. When manure is removed from storage by the Crop and Soil module, emptied manure outputs report the quantity of manure and nutrients removed on that day, and Manure Stream attributes representing stored manure are updated accordingly to reflect post-removal amounts remaining in storage.

D. Manure composition updates

Received manure

In composting processors, the following nutrient sources are represented in received manure values: ManureStream values, as received from the previous processor(s) in the manure management chain. For composting processors, these ManureStream instance(s) typically represent an accumulated quantity of manure from an open lot or bedded pack, daily/weekly cleanout of the manure/bedding mix generated by other pen types, or daily addition of separated manure solids. Daily precipitation volume/mass is not currently represented in received manure added to compost (i.e., precipitation falling each day is not represented in the received manure outputs representing nutrient/mass/water additions to compost each day).

Stored manure

Below is a summary of updates to ManureStream variables representing the stored manure. Note that the formulas below may be a summarization of multiple steps detailed above, and are intended to provide an overview of what mass losses/gains are reflected in the value of each variable. Equations in the table below (Calculation column) are in the format of: updated stored manure value = yesterday's stored manure value + today's manure value +/- XYZ. The updated stored manure values reflect the total quantity of manure/nutrients in storage on a single day after accounting for all gains/losses that occurred on that day.

Table 36: Manure storage variable calculations

Variable	Units	Calculation
water	kg	Stored manure water (kg) + received manure water (kg)
total_ammoniacal_nitrogen	kg	max(0, stored ammoniacal nitrogen (kg) + received ammoniacal nitrogen (kg) − NH_3N emissions (kg))
nitrogen	kg	stored nitrogen (kg) + received nitrogen (kg) − NH_3-N emissions (kg) − N_2O-N emissions (kg)
phosphorus	kg	stored phosphorus (kg) + received phosphorus (kg)
potassium	kg	stored potassium (kg) + received potassium (kg)
ash	kg	stored ash (kg) + received ash (kg)
degradable_volatile_solids		stored degradable VS (kg)+received degradable VS (kg)− VSd loss (kg)
manure_non_degradable_volatile_solids	kg	stored manure non-degradable VS (kg) + received manure non-degradable VS (kg) − VSnd loss (kg)

Continued on next page

Variable	Units	Calculation
bedding_non_degradable_volatile_solids	kg	stored bedding non-degradable VS (kg) + received bedding non-degradable VS (kg)
total_solids	kg	stored total solids (kg) + received total solids (kg) − total volatile solids loss (kg)
volume	m^3	stored volume (m^3) + received volume (m^3) − $\frac{\text{total volatile solids loss (kg)}}{\text{SOLID\_MANURE\_DENSITY}}$

[1] The "Max(0,) notation prevents the ammoniacal N value from becoming negative. This is especially important early in a simulation when accumulated manure quantities that are very small compared to the fixed surface area value can lead to high ammonia emissions.

[2] degradable_volatile_solids_frac = received manure degradable_volatile_solids / received manure total_volatile_solids

XI. Daily spread

A. Introduction

Daily spread is a method of manure management in which manure is collected daily (or several times per week) directly from animal housing areas and field-applied. This manure management strategy may be more labor intensive but can greatly reduce the quantity of manure a farm must store.

Implementation in RuFaS

Daily spread in RuFaS is implemented as essentially a "blank" manure processor. It receives manure but does not estimate any gas emissions, nutrient losses, or changes to manure composition. **Note that this section covers strictly the Manure Module daily spread functionality; the actual daily application of manure from the manure module occurs in the Crop and Soil module and is thus covered in the Crop and Soil documentation.**

Classes

Table 37: List of classes for daily spread.

Class	Description
DailySpread(Storage)	daily_spread.py

Required User Inputs

Table 38: Required inputs for the daily spread section (refreshed_manure_management.json)

Variable	Definition (units)	Description
Name	none	Unique identifier of the specific daily spread configuration used.

Other inputs

Instance(s) of ManureStream for each manure stream defined by the user that represent the attributes of the manure in the specific manure stream. ManureStream instances include the following variables (all in kg except for volume, m^3 and manure methane production potential, $m^3/kgVS$):

- water
- ammoniacal_nitrogen
- nitrogen
- phosphorus
- potassium
- ash
- manure_degradable_volatile_solids
- manure_non_degradable_volatile_solids
- bedding_non_degradable_volatile_solids
- total_solids

- mass (equal to sum of water and total solids)
- total volatile solids (equal to sum of degradable and non-degradable volatile solids)
- volume
- methane_production_potential

Expected Outputs

- ManureStream variables representing manure received by daily spread processor each day

XII. Manure to Soil and Crop Module Connection

A. Introduction

On most dairies, the amount of manure applied to a field is determined by the amount of nitrogen (N) and phosphorus (P) needed by the crop to support production, or by the maximum amount of that nutrient that can be applied based on environmental regulations. These amounts are often established in the farm's detailed nutrient management plan and updated with crop yields and manure quantities and compositions.

Often, application of manure alone cannot fulfill all crop nutrient needs due to a mismatch between the N:P ratio required by the crop and the N:P ratio of the available manure. The manure application is therefore referred to as being **limited** by the nutrient whose requirement is met first through manure application.

In RuFaS, the user provides the desired amount of manure N and P for each manure application and specifies whether the model should supplement the application with imported manure or fertilizer if the exact amounts of both N and P are not met by the manure available on-farm. If both N and P cannot be supplied by manure in storage on the day of application, the total amount of the limiting nutrient (N or P) is used to determine the amount of the other nutrient, the total manure mass, and other accompanying nutrients (e.g., volatile solids (VS), potassium (K)) based on manure composition. This information is then passed to the Soil and Crop Module for application of nutrients to fields and to the Manure Module for removal of stored nutrients.

B. Implementation in RuFaS

In RuFaS, manure accumulates in storage until it is removed for field application, transferred to another storage, or exported off-farm. This accumulated manure represents the manure available for field application. Manure applications, referred to as *manure nutrient requests*, originate from the Crop and Soil Module and are scheduled by year and Julian day. Each request contains a specified quantity of manure N and P.

Nutrient requests may be made for either solid or liquid manure. Manure is designated as liquid or solid based on storage type: slurry storage and anaerobic lagoons contain liquid manure, while all other storage types contain solid manure.

Exchange of the information about the manure nutrient application request and the availability of manure nutrients is executed in the simulation engine to ensure independence of the Manure Module and the Soil and Crop Module. The flow of information between the Manure and Soil and Crop Modules for a manure application is illustrated in Figure 4.

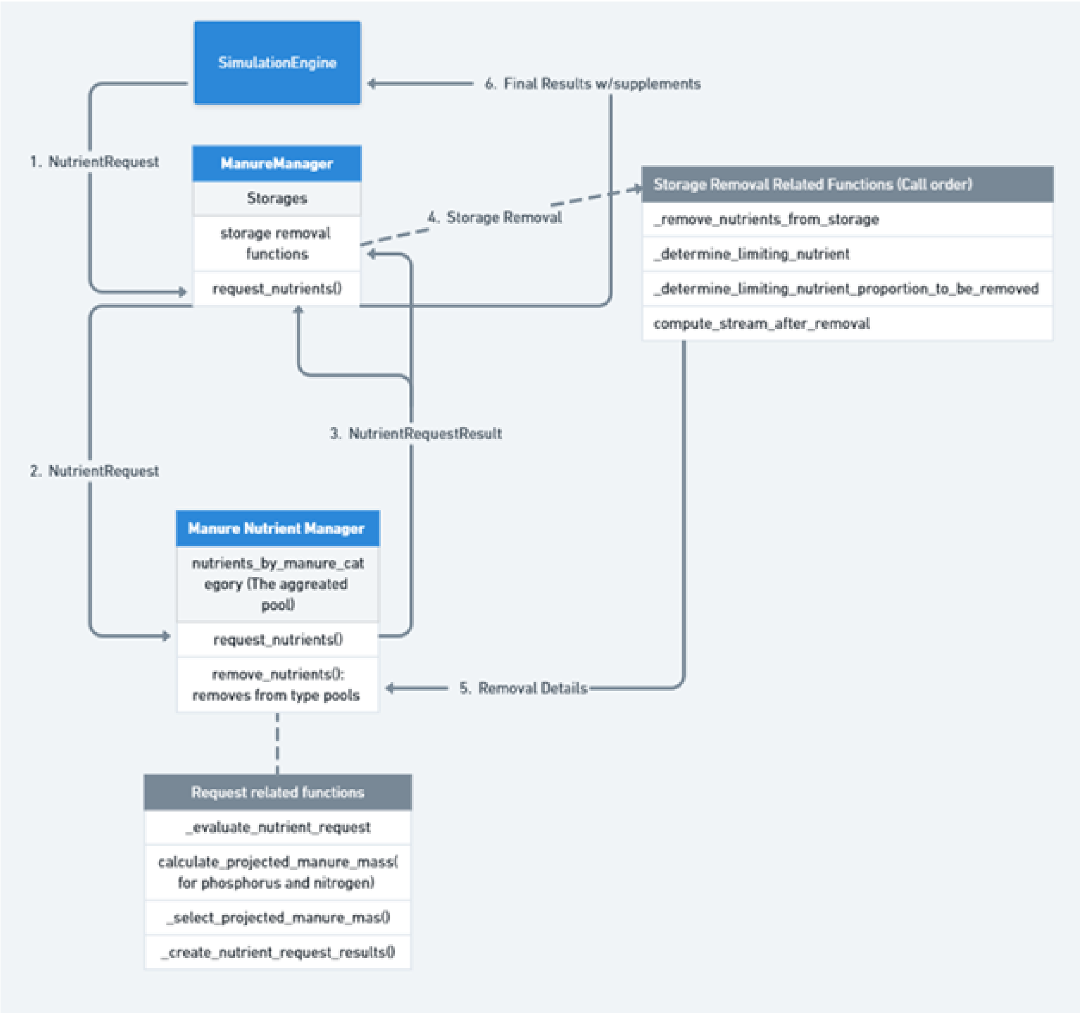

Figure 4

Each day, the SimulationEngine calls the generate_daily_manure_applications() method to check with the FieldManager to see if there is a manure application scheduled and, when there is, pass the manure nutrient request created by the FieldManager to the ManureManager. The ManureManager, in turn, checks with the ManureNutrientManager to see if the manure nutrient request can be fulfilled via the request_nutrients() in the ManureManager, which calls the evaluate_nutrient_request() function from the ManureNutrientManager class. The ManureNutrientManager then evaluates the nutrient request, determines the limiting nutrient to be used, and calculates the amount of each manure nutrient and the total mass that can be provided by the manure in storage using the following logic. At this time, nutrients from all storages are pooled by ManureNutrientManager to determine the total quantity of nutrients available. Once the nutrient request is returned, an identical proportion (not amount) of manure is removed from all storages. In other words, the user cannot specify at this time which storage(s) should be pulled from for a manure application.

C. Methodology

Step 1: Determination of the Limiting Nutrient

Determine the nutrient that manure storage removal will be based on using the composition of N and P in the manure nutrient pool. This step is executed by evaluate_nutrient_request:

$$\text{manure\_mass\_N} = \frac{\text{mass\_N\_requested}}{\text{manure\_nitrogen\_frac}} \tag{1}$$

$$\tag{2}$$

$$\text{manure\_mass\_P} = \frac{\text{mass\_P\_requested}}{\text{manure\_phosphorus\_frac}} \tag{3}$$

The projected manure mass is then calculated as:

$$\text{projected manure mass} = \min\left(\text{mass\_N\_requested}, \text{mass\_P\_requested}\right) \tag{4}$$

[MN.MSC.1]

where:

- mass_N_requested is the mass of manure (kg) required to meet the requested mass of N.
- mass_P_requested is the mass of manure (kg) required to meet the requested mass of P.

The limiting nutrient is the nutrient (N or P) with a smaller projected mass required to meet the nutrient request. The projected manure mass based on the limiting nutrient is used to determine the removal of manure from the manure storages.

Step 2: Proportion of Limiting Nutrient Removed

Using the limiting nutrient identified in Step 1 and considering the total quantity of that nutrient available in the aggregated ManureNutrient pool, the proportion of the limiting nutrient to be removed from each storage is calculated as:

$$\text{nutrient\_proportion\_to\_be\_removed} = \min\left(\frac{\text{mass\_requested\_nutrient}}{\text{manure\_nutrient\_available}}, 1\right) \tag{5}$$

[MN.MSC.2]

Step 3: Removal of Nutrients from Individual Storages

Next, the amount of the limiting nutrient and projected mass of manure to be removed from the aggregated ManureNutrient pool are determined. If there is sufficient manure available to fulfill the nutrient request, the quantity of the limiting nutrient removed will be equal to the requested amount; otherwise, the amount removed will be lower than the requested amount.

For each nutrient and each storage, the nutrient removal is calculated as:

$$\text{nutrient\_to\_be\_removed\_from\_storage} = \text{nutrient\_in\_storage} \times \text{nutrient\_proportion\_to\_be\_removed} \tag{6}$$

[MN.MSC.3]

where:

- nutrient_to_be_removed_from_storage is the mass of the nutrient removed from a specific manure storage (kg).
- nutrient_in_storage is the mass of the nutrient in the storage prior to manure application (kg).

The ManureNutrientManager then passes the result of the manure nutrient request with the total amount of the limiting nutrient to be supplied and the ManureManager calculates the mass of manure and nutrients to be removed from each individual storage. The ManureManager returns the final results of the manure application to the SimulationEngine, including:

- Total nitrogen mass supplied (kg)
- Total phosphorus mass supplied (kg)
- Total manure mass supplied (kg)
- Total dry matter supplied (kg)
- Dry matter fraction of the manure (%)
- A Boolean indicator of whether the nutrient request was fulfilled

After the nutrient request is partially or fully fulfilled, the corresponding quantities of mass and nutrients are subtracted from each individual manure storage.

XIII. Key Constants

Table 39: The key constants for all equations in the Manure Module

Variable	Value	Definition
MANURE_DAMPING_FACTOR	0.65	Fixed damping factor applied to the air temperature amplitude.
MANURE_TEMPERATURE_LAG	30	Lag constant representing delayed thermal response of manure temperature relative to air temperature (days).
ANAEROBIC_LAGOON_MANURE_-RETENTION	0.1	Fraction of stored manure retained in anaerobic lagoon when storage interval is reached.
ACTIVATION_ENERGY	81,000 J/mol	Apparent activation energy of methanogenesis in dairy manure.
DEFAULT_STORED_MANURE_PH	7.5	Default pH of manure in slurry storage or anaerobic lagoon.
FREEBOARD_CONSTANT	1.2	20% volume allowance above max volume if surface area not user-defined.
DEPTH_CONSTANT	4.572	Depth of slurry/liquid manure storage used for surface area calculation (m).
PRECIPITATION_CONSTANT	0.25	Annual precipitation constant for determining storage surface area (m).
MANURE_CONVERSION_CONSTANT	0.1175	Factor to estimate manure volume per cow per day (m^3).
METHANE_DESTRUCTION_EFFICIENCY	0.81	Percent methane destroyed with cover and flare system.
VS_TO_METHANE_LOSS_RATIO	6.665 kg	Mass ratio of CO_2+CH_4 to CH_4 from storage (Petersen et al., 2024).
NATURAL_LOG_ARRHENIUS_CONSTANT	30.6 g CH_4/kg VS/h	Log of Arrhenius parameter for methane emissions from stored manure.
STORAGE_RESISTANCE	23.1	Default resistance to ammonia volatilization (s/m).
SLURRY_MANURE_DENSITY	990 kg/m^3	Default density of manure as excreted.
AMMONIA_EMISSION_COEFFICIENT_-UNTILLED	0.25	Ammonia emission coefficient (no mixing).
AMMONIA_EMISSION_COEFFICIENT_-TILLED	0.5	Ammonia emission coefficient (with mixing).

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Variable	Value	Definition
DEFAULT_DAYS_SINCE_LAST_MIXING	1 d	Days since last mixing, for C decomposition.
NITROUS_OXIDE_EMISSION_UNTILLED	0.01	N_2O-N emitted per kg manure N/day (no mixing).
NITROUS_OXIDE_EMISSION_TILLED	0.07	N_2O-N emitted per kg manure N/day (with mixing).
HOUSING_SPECIFIC_CONSTANT	260.0 s/m	Default constant for ammonia emissions from housing.
DEFAULT_PH_FOR_HOUSING_AMMONIA	7.7	Default pH for manure on housing floors.
MILKING_FRESH_WATER_USE_RATE	30 L/animal/day	Milking water use rate per animal.
WATER_DENSITY_KG_PER_M3	0.997 kg/m^3	Default water density.
CARBON_DIOXIDE_MOLAR_MASS	44.01 g/mol	Molar mass of CO_2.
CARBON_DIOXIDE_TO_METHANE_RATIO	4–6	Volumetric ratio of CO_2 to CH_4 during digestion.
IDEAL_GAS_LAW_R	0.0821 L atm/mol K	Ideal gas law constant.
METHANE_MOLAR_MASS	16.04 g/mol	Molar mass of CH_4.
TAN_INCREASE_FACTOR	1.60	TAN increase from anaerobic digestion (unitless).
DEFAULT_LAYER_TEMPERATURE	30 °C	Default layer temperature for decomposition.
DECOMPOSITION_TEMPERATURE	60 °C	Temperature for peak microbial activity.
DEFAULT_CARBON_FRACTION_VSD	0.5	Carbon content of degradable VS.
DEFAULT_CARBON_FRACTION_VSND	0.35	Carbon content of non-degradable VS.
DEFAULT_MOISTURE_EFFECT	0.65	Moisture effect on microbial decomposition.
DEFAULT_LAG_TIME	2 d	Lag time for C decomposition.
EFFECTIVE_MICROBIAL_DECOMP_RATE	0.00237	Microbial decomposition rate (unitless).
FIRST_ORDER_DECAY_COEFFICIENT	0.01	First-order decay coefficient.
LEACHING_COEFFICIENT	–	N leached per kg manure N/day.
DEFAULT_DAYS_SINCE_LAST_HARROW	1 d	Days since last harrow event.
ACHIEVABLE_METHANE_EMISSION	0.24 m^3 CH_4/kg VS	Achievable methane generation.
SOLID_MANURE_DENSITY	700 kg/m^3	Default solid manure density.

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Variable	Value	Definition
LIQUID_MANURE_DENSITY	1000 kg/m^3	Default liquid manure density.

XIV. References

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